

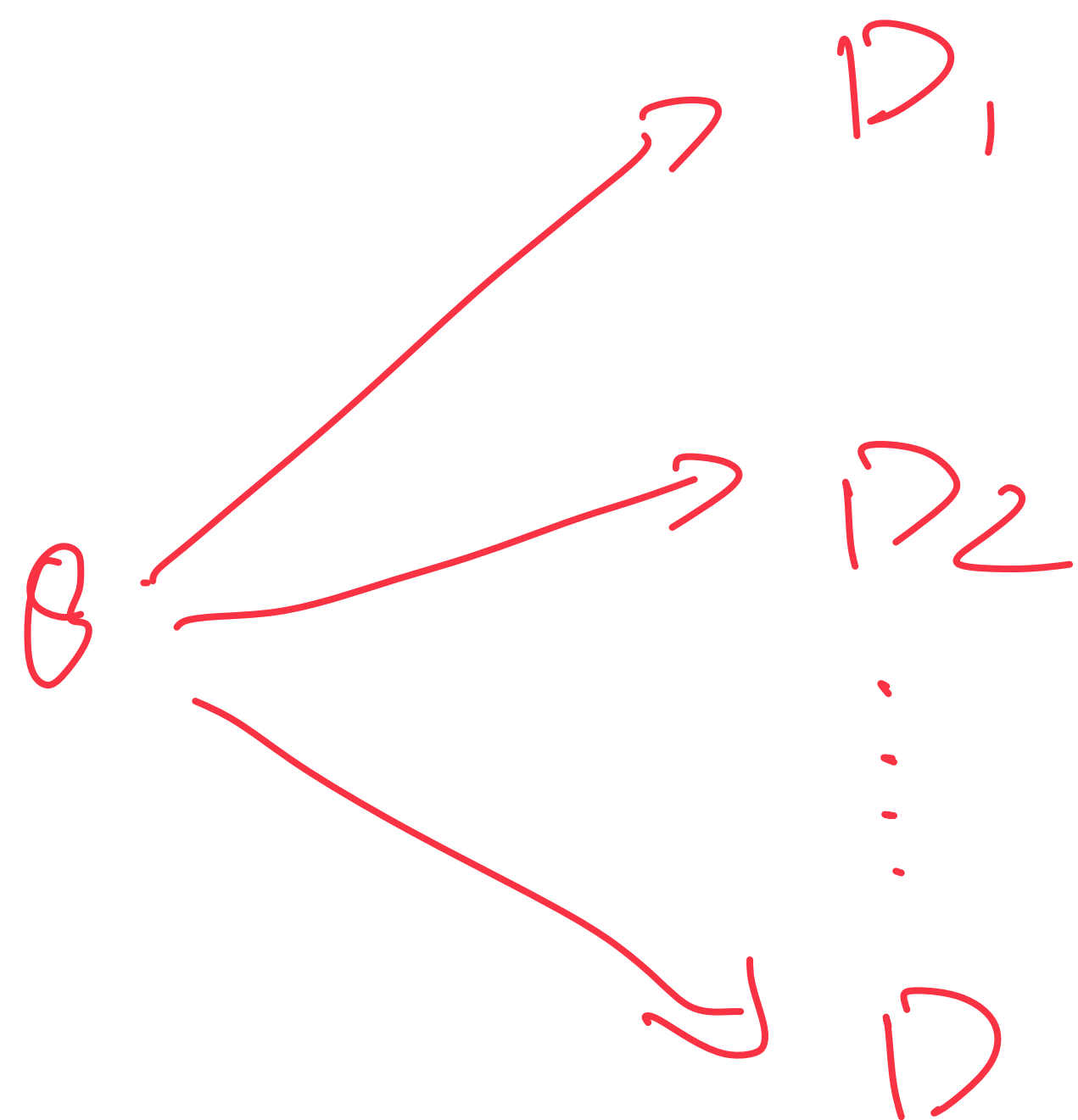
STAT 420/812

Computational Statistics

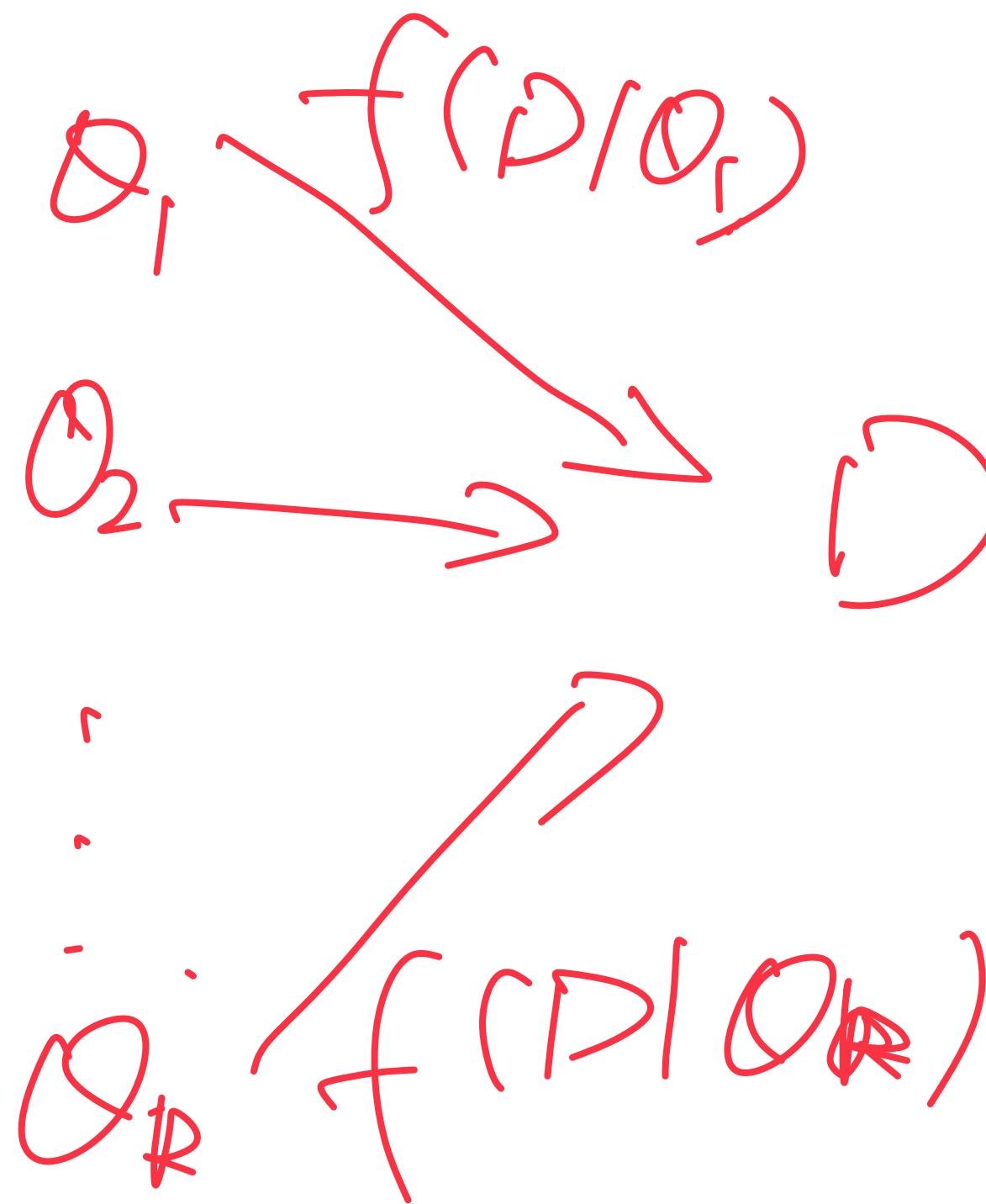
Maximum Likelihood Estimate

1; ~~defined~~

$$f(D|Q)$$



$$L(Q; D) = f(\underset{\Delta}{D}|Q)$$



Maximum Likelihood Estimator

$$\hat{\theta}_{MLE} = \underset{\theta}{\operatorname{argmax}} L(\theta; D)$$

Example:

$$p(y_i | \theta) = \theta^{y_i} (1-\theta)^{1-y_i}$$

$y_1, y_2, y_3 | \theta \stackrel{iid}{\sim} \text{Bern}(\theta)$, θ is pop. prop. $\in [0, 1]$

$$y_1 = 1, y_2 = 1, y_3 = 0$$

$$P(y_1=1, y_2=1, y_3=0 | \theta) = \prod_{i=1}^3 p(y_i | \theta)$$

θ	$P(y_1=1, y_2=1, y_3=0 \theta) = \prod_{i=1}^3 p(y_i \theta)$			
0	0	0	1	= 0
$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	= $\frac{1}{8} = 0.1250$!
<u>$\frac{2}{3}$</u>	$\frac{2}{3}$	$\frac{2}{3}$	$\frac{1}{3}$	= $\frac{4}{27} = 0.1481$.
$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{1}{4}$	= $\frac{9}{64} = 0.1406$.
1	1	1	0	= 0

$$P(y_i | \theta) = \theta^{y_i} (1-\theta)^{1-y_i} \leftarrow$$

$$L(\theta | y_1, \dots, y_n) = \prod_{i=1}^n P(y_i | \theta)$$

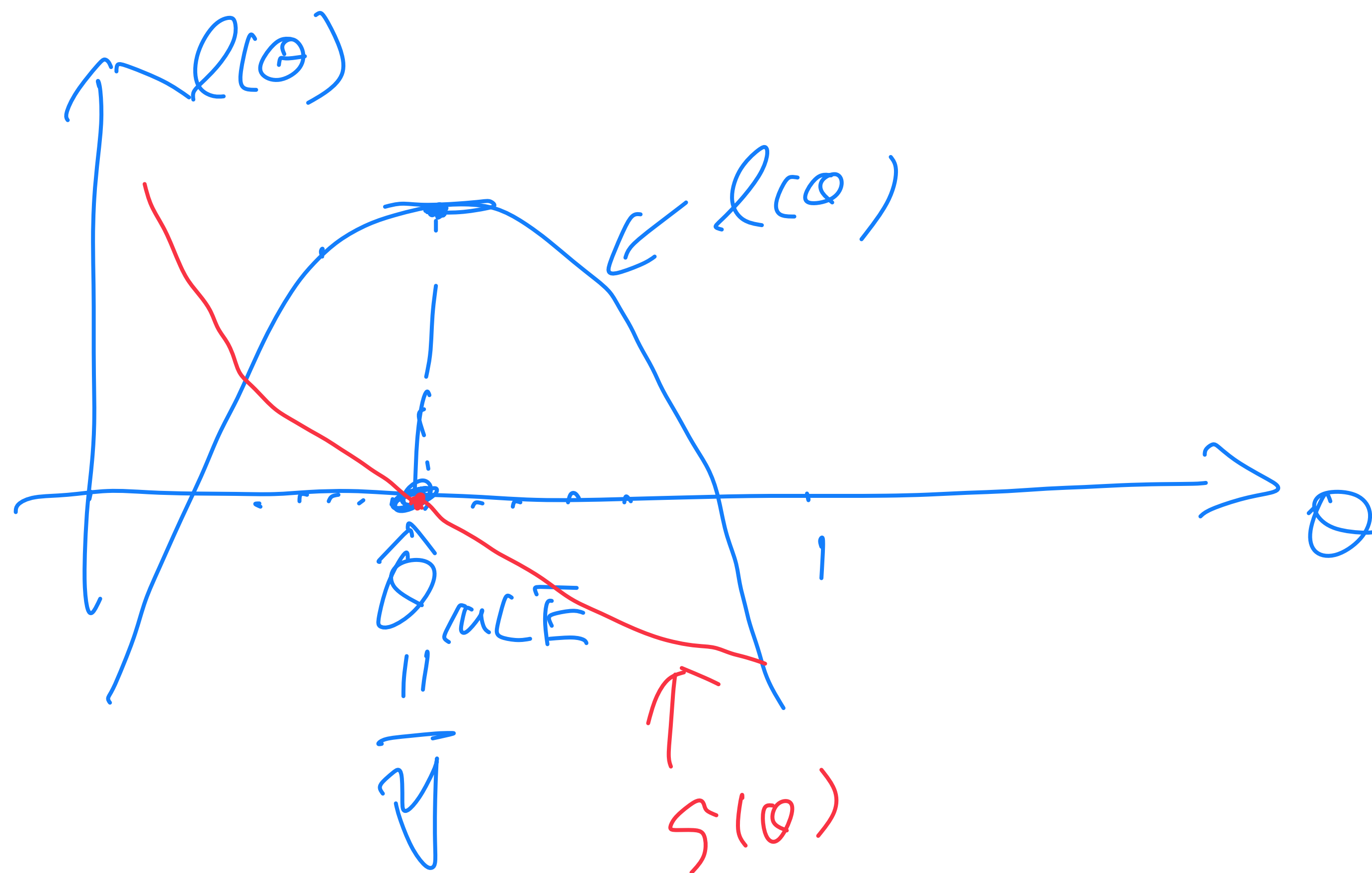
$$= \prod_{i=1}^n \theta^{y_i} (1-\theta)^{1-y_i}$$

$$= \theta^{\sum_{i=1}^n y_i} (1-\theta)^{n - \sum_{i=1}^n y_i}$$

$$= \theta^{n_1} (1-\theta)^{n_0}$$

$$\ell(\theta) = \log \frac{L(\theta)}{\Delta}$$

$$= n_1 \log \theta + n_0 \log(1-\theta)$$



Score function

$$S(\theta) = l'(\theta) \equiv 0$$

$$l(\theta) = n_1 \log \theta + n_0 \log(1-\theta)$$

$$S(\theta) = \frac{n_1}{\theta} - \frac{n_0}{1-\theta} \equiv 0$$

$$\begin{aligned} \frac{n_1}{\theta} &= \frac{n_0}{1-\theta} \Leftrightarrow \frac{\theta}{1-\theta} = \frac{n_1}{n_0} \\ \Leftrightarrow \frac{\theta}{1} &= \frac{n_1}{n} = \frac{\sum_{i=1}^n y_i}{n} = \bar{y} \end{aligned}$$

$$\text{where } n = n_1 + n_0$$

Example

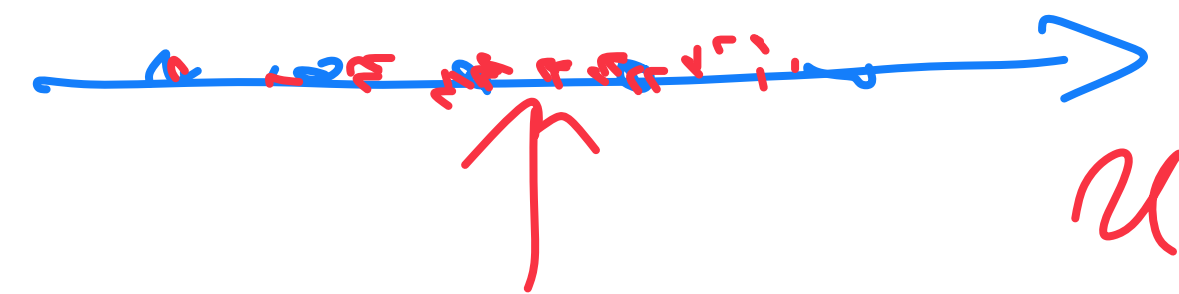
$$x_1, \dots, x_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma_0^2)$$

$$f(x_i | \mu) = \frac{1}{\sqrt{2\pi} \sigma_0} e^{-\frac{(x_i - \mu)^2}{2\sigma_0^2}}$$

$$l(\mu; x_1, \dots, x_n) = \sum_{i=1}^n \log f(x_i | \mu)$$

$$= \sum_{i=1}^n \left[\log\left(\frac{1}{\sqrt{2\pi} \sigma_0}\right) - \frac{(x_i - \mu)^2}{2\sigma_0^2} \right]$$

$$= n \cdot \log\left(\frac{1}{\sqrt{2\pi} \sigma_0}\right) - \frac{\sum_{i=1}^n (x_i - \mu)^2}{2\sigma_0^2}$$

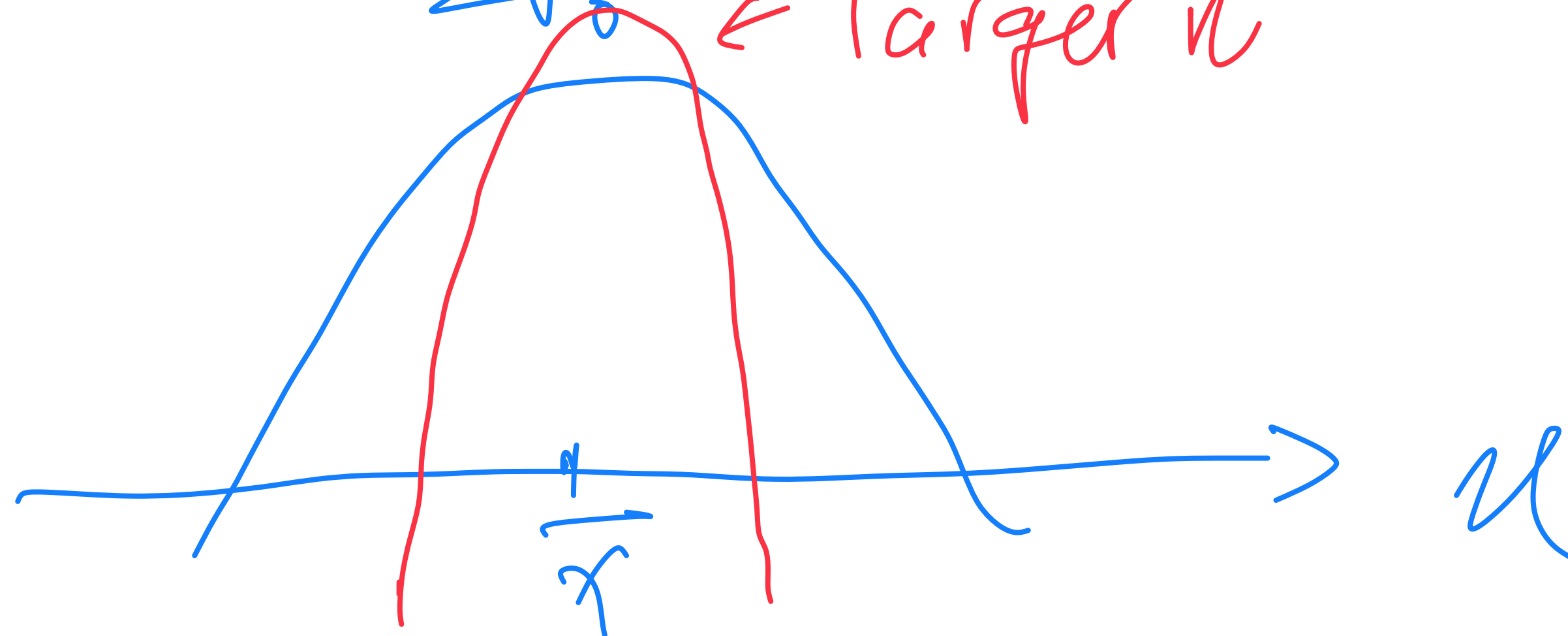


$$= C - \frac{1}{2\sigma_0^2} \sum_{i=1}^n (\mu - x_i)^2$$

$$= C - \frac{1}{2\sigma_0^2} \cdot \sum_{i=1}^n \left[(\mu - \bar{x})^2 + (\bar{x} - x_i)^2 + 2(\mu - \bar{x})(\bar{x} - x_i) \right]$$

$$= C_2 - \frac{n}{2\sigma_0^2} (\mu - \bar{x})^2$$

← larger n



Score function

$$S(\mu; x_1, \dots, x_n) = \frac{\partial}{\partial \mu} \ell(\mu; x_1, \dots, x_n)$$

$$= 0 - \frac{\sum_{i=1}^n z(\mu - x_i)}{\sigma_0^2}$$

$$= - \frac{n\mu - \sum x_i}{\sigma_0^2} = - \frac{n}{\sigma_0^2} (\mu - \bar{x})$$

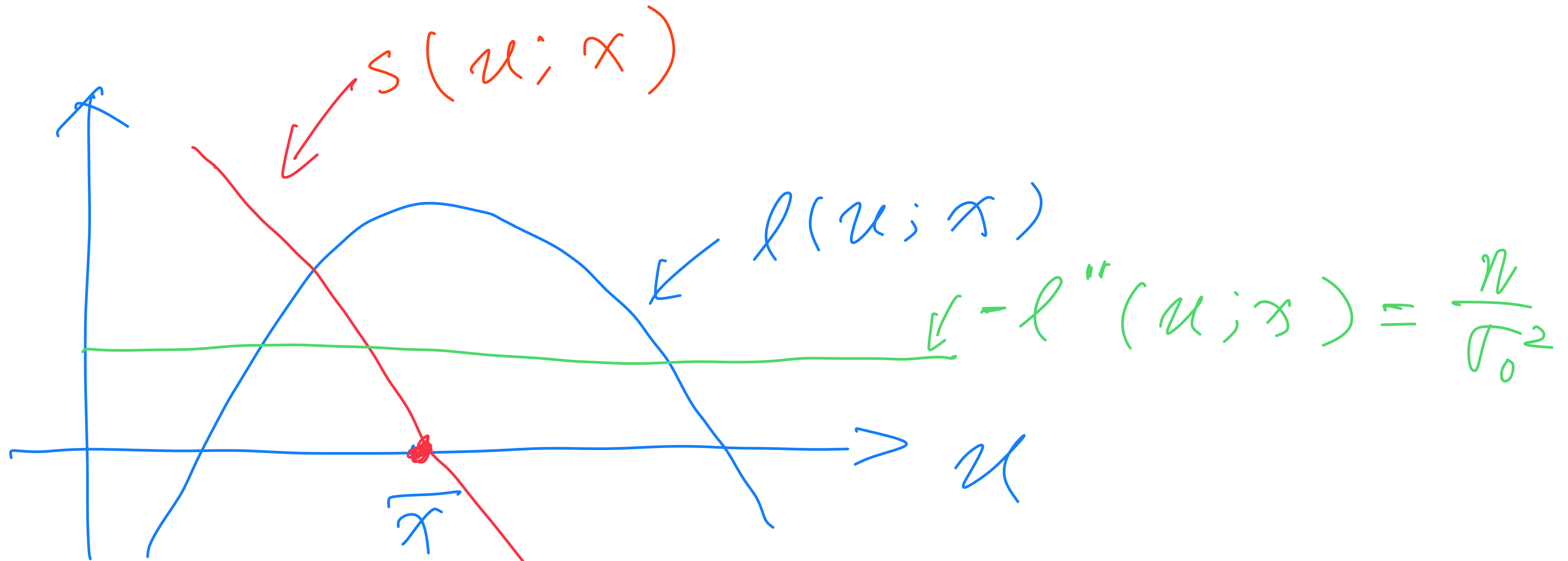
$$-\ell''(\mu; x) = \frac{n}{\sigma_0^2} \leftarrow \text{Fisher information}$$

$$S(\mu; x_1, \dots, x_n) \equiv 0$$

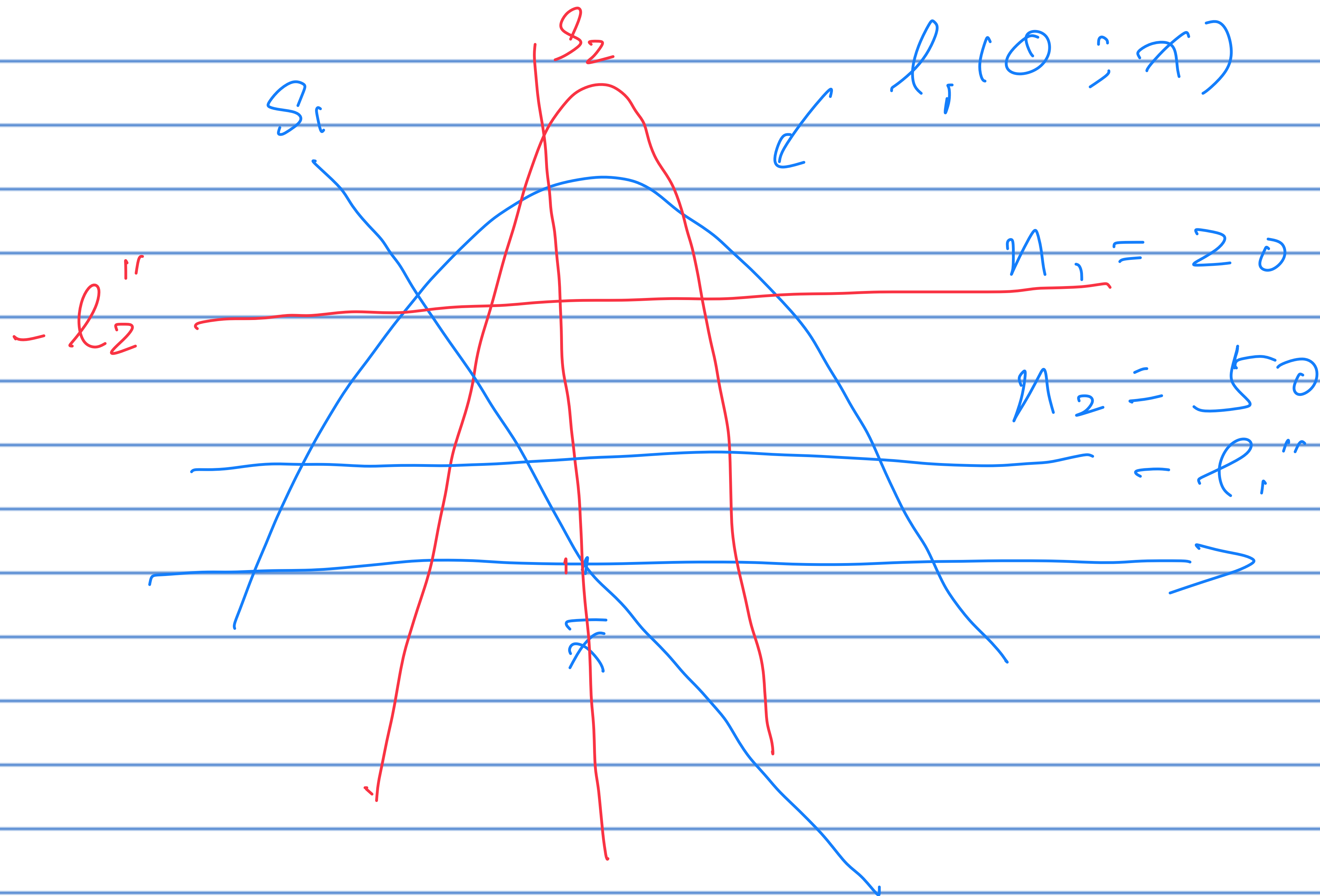
$$n\mu - \sum x_i = 0$$

$$\Leftrightarrow \mu = \frac{\sum x_i}{n} = \bar{x}$$

$$\hat{\mu}_{MLE} = \bar{x}$$

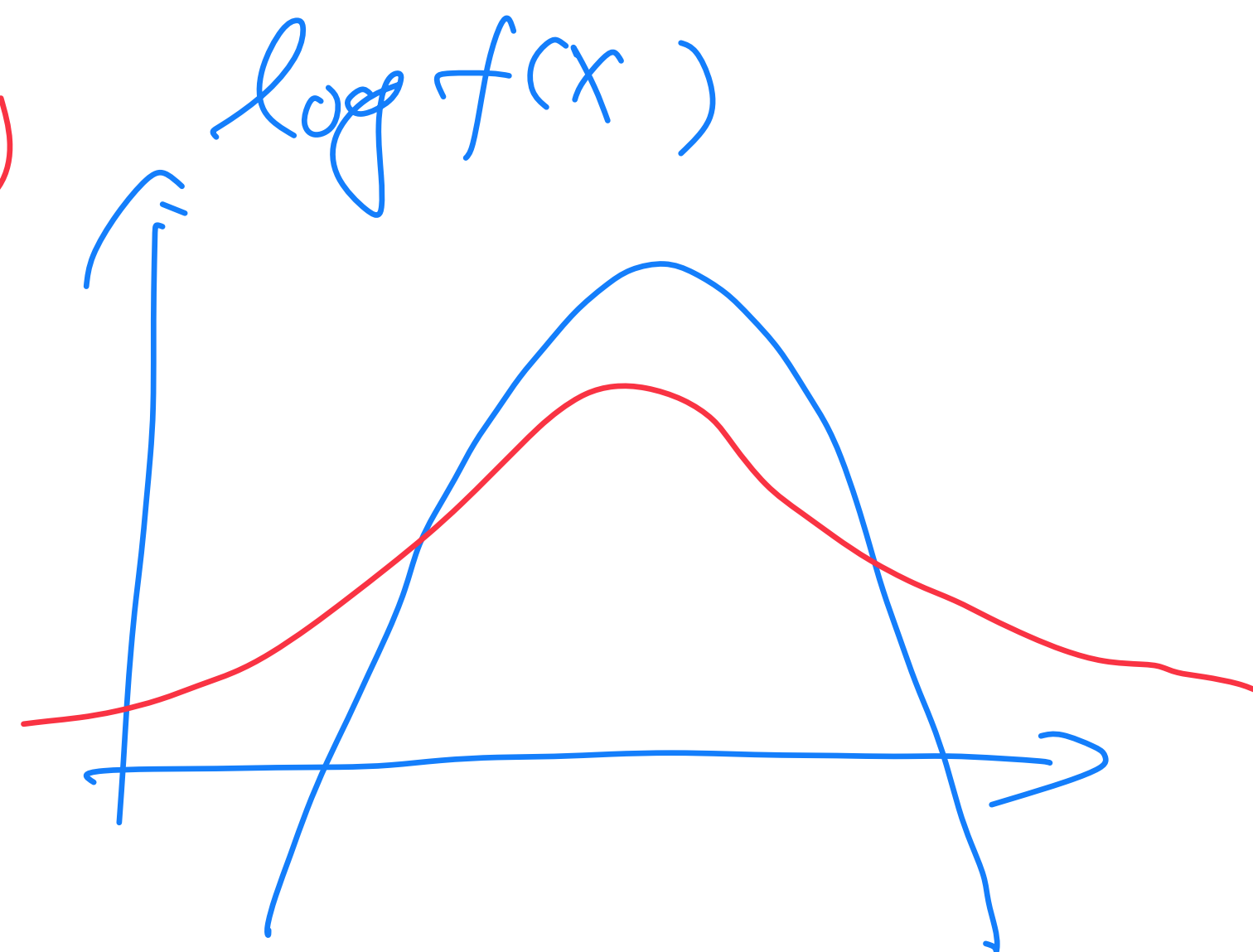


log-likelihood, score, Hessian function



Example (Cauchy model)

$x_1, \dots, x_n \sim \text{Cauchy}(0)$



$$f(x; 0) = \frac{1}{\pi} \frac{1}{1 + (x - 0)^2}, \quad x = (x_1, \dots, x_n)$$

$$\Rightarrow \ell(0; x) = -(\log(\pi)) \cdot n - n \sum_{i=1}^n \log(1 + (x_i - 0)^2)$$

$$\Rightarrow S(0; x) = -n \sum_{i=1}^n \left[\frac{1}{1 + (x_i - 0)^2} \cdot (0 - x_i) \right]$$

properties of MLE.

Under some regularity conditions (satisfied by most statistical models used in practice), the MLE θ_{mle} for iid samples has the following properties:

1. The MLE is consistent: $\hat{\theta}_{\text{mle}} \rightarrow \theta_*$ when $n \rightarrow \infty$

2. The MLE is asymptotically normal:

$$\hat{\theta}_{\text{mle}} \rightarrow N(\theta_*, (n I(\theta_*))^{-1})$$

where

$$I(\theta_*) = -E \left(\frac{\Delta^2}{\Delta \theta^2} \log(f(X|\theta)) \Big|_{\theta_*} \right) = E \left(\left(\frac{d^2}{d\theta_i d\theta_j} \log(f(X|\theta)) \right)_{p \times p} \right) \quad (7.10)$$

3. Plug in $\hat{\theta}_{\text{mle}}$ in $I(\theta_*)$, the fact still holds:

$$\hat{\theta}_{\text{mle}} \rightarrow N(\theta_*, (n I(\hat{\theta}_{\text{mle}}))^{-1}) \quad (7.11)$$

$I(\hat{\theta}_{\text{mle}})$ is called the expected fisher information. In contrast, $I(\theta_*)$ can also be estimated by:

$$-\frac{1}{n} \sum_i \frac{\Delta^2}{\Delta \theta^2} \log(f(X_i|\theta)) \Big|_{\hat{\theta}_{\text{mle}}} \cdot n = -\frac{\partial^2}{\partial \theta^2} \ell(\theta; X) \quad (7.12)$$

This is called **the observed fisher information**, ~~n times of which is just the minus of the second derivative of the log likelihood function at $\hat{\theta}_{\text{mle}}$~~ . The obvious advantage of this estimation of standard deviation is that we do not need to compute the expected value in (7.10), which might be difficult or impossible.

4. The MLE is asymptotically optimal.

$$\begin{pmatrix} \frac{\partial^2 f}{\partial \theta_1 \partial \theta_1} & \dots & \frac{\partial^2 f}{\partial \theta_1 \partial \theta_p} \\ \vdots & & \vdots \\ \frac{\partial^2 f}{\partial \theta_p \partial \theta_1} & \dots & \frac{\partial^2 f}{\partial \theta_p \partial \theta_p} \end{pmatrix} \quad (7.9)$$



E w.r.t. X

$n \cdot I(\theta_*)$

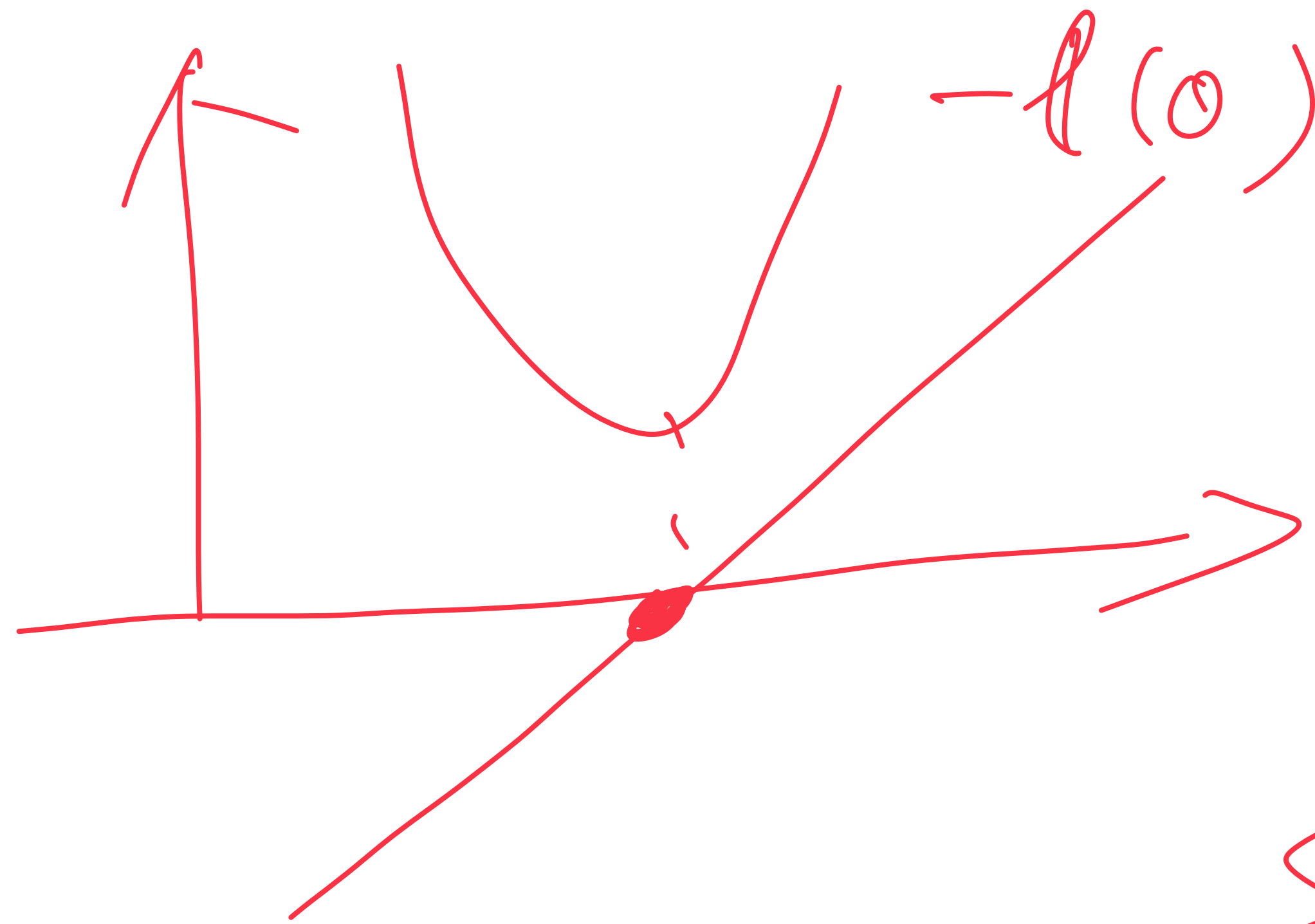
$$E[f(X)] \approx \frac{f(x_1) + \dots + f(x_n)}{n}$$

Suppose $X_1, \dots, X_n$ iid $f(x_i | \theta)$

$$l(\theta | X_1, \dots, X_n) = \sum_{i=1}^n \log f(x_i | \theta)$$

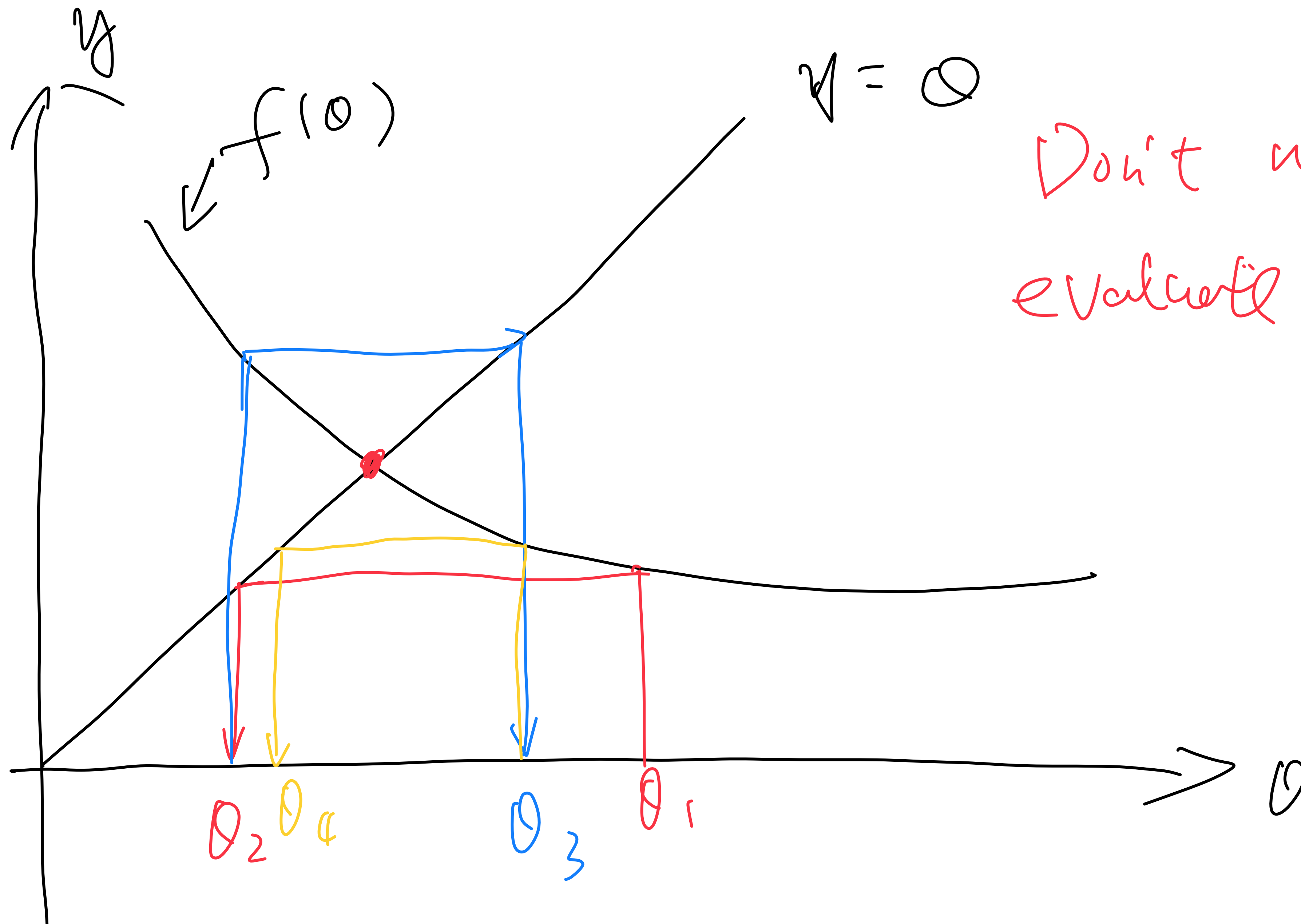
$$-l''(\theta | X_1, \dots, X_n) =$$

Simple iteration



$$S(0) = f'(0)$$

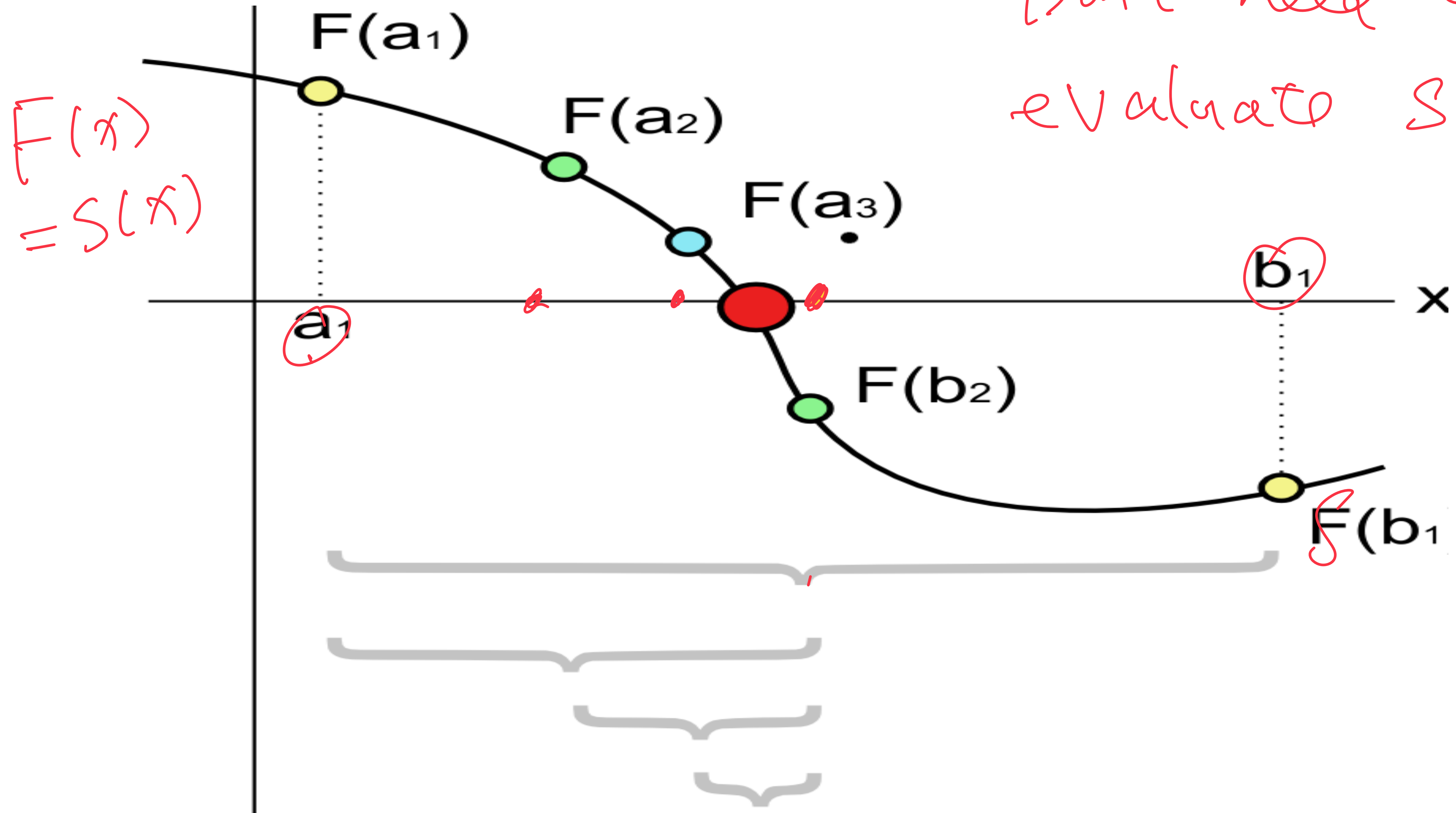
Finding solution to $S(0) = 0 \Leftrightarrow 0 = f(0)$



Don't need to
evaluate $f'(\theta)$

Bisection method.

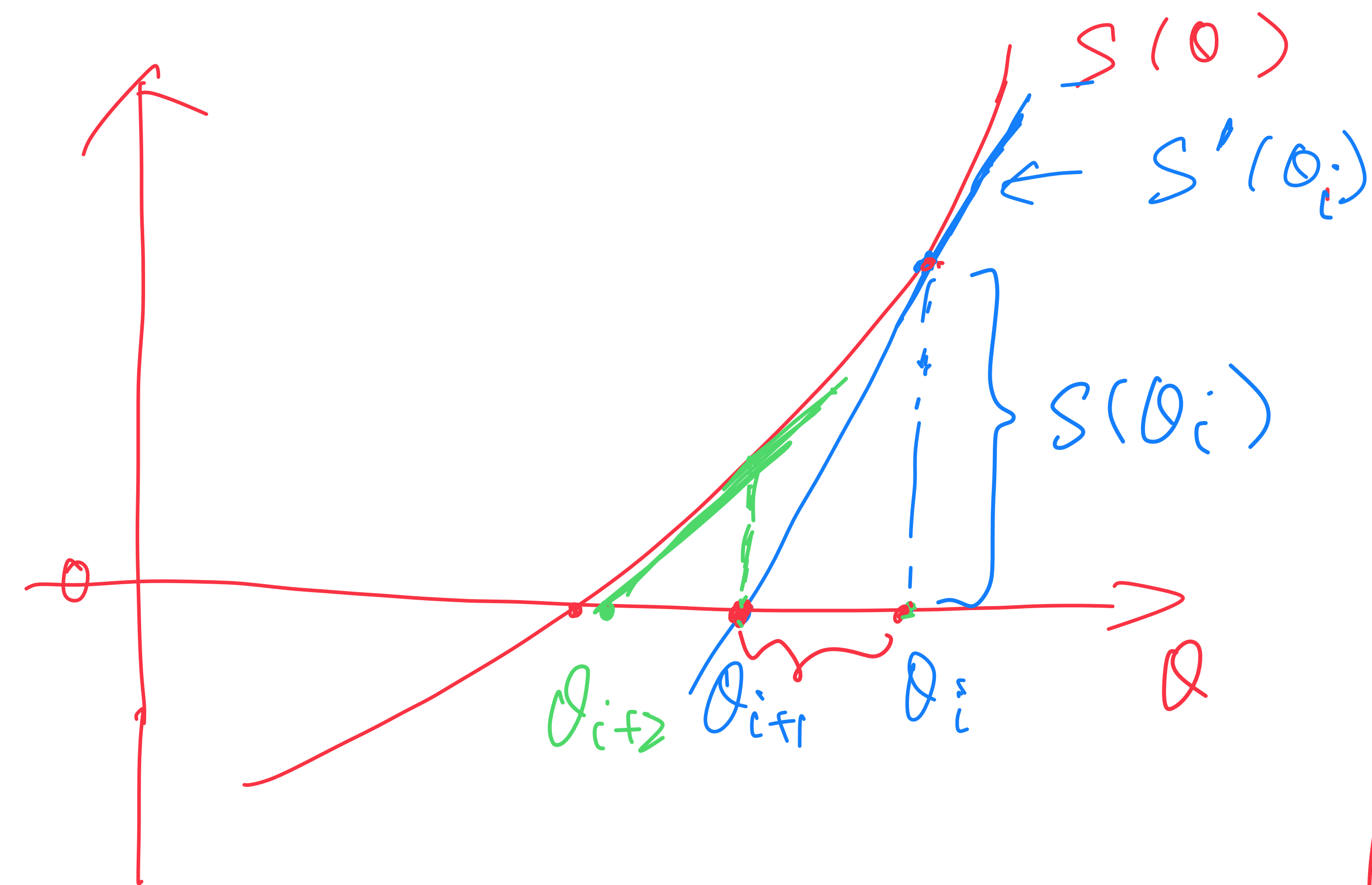
Don't need to
evaluate $S'(0)$



Newton-Raphson

$$S(\theta) = -\ell'(\theta; D), S(\hat{\theta}_{MLE}) = 0$$

$$S'(\theta) = -\ell''(\theta; D)$$



$$\frac{S(\theta_i) - 0}{\theta_i - \theta_{i+1}} = S'(\theta_i)$$

$$\frac{\theta_i - \theta_{i+1}}{S(\theta_i)} = \frac{1}{S'(\theta_i)}$$

$$\theta_i - \theta_{i+1} = \frac{S(\theta_i)}{S'(\theta_i)}$$

$$\theta_{i+1} = \theta_i - \frac{S(\theta_i)}{S'(\theta_i)} \leftarrow \frac{-\ell'(\theta_i)}{-\ell''(\theta_i)}$$

step size

Illustration of NR

$$f(x) = a(x-u)^2 + c$$

$$u = \arg\min f(x)$$

$$f'(x) = 2a(x-u)$$

$$f''(x) = 2a$$

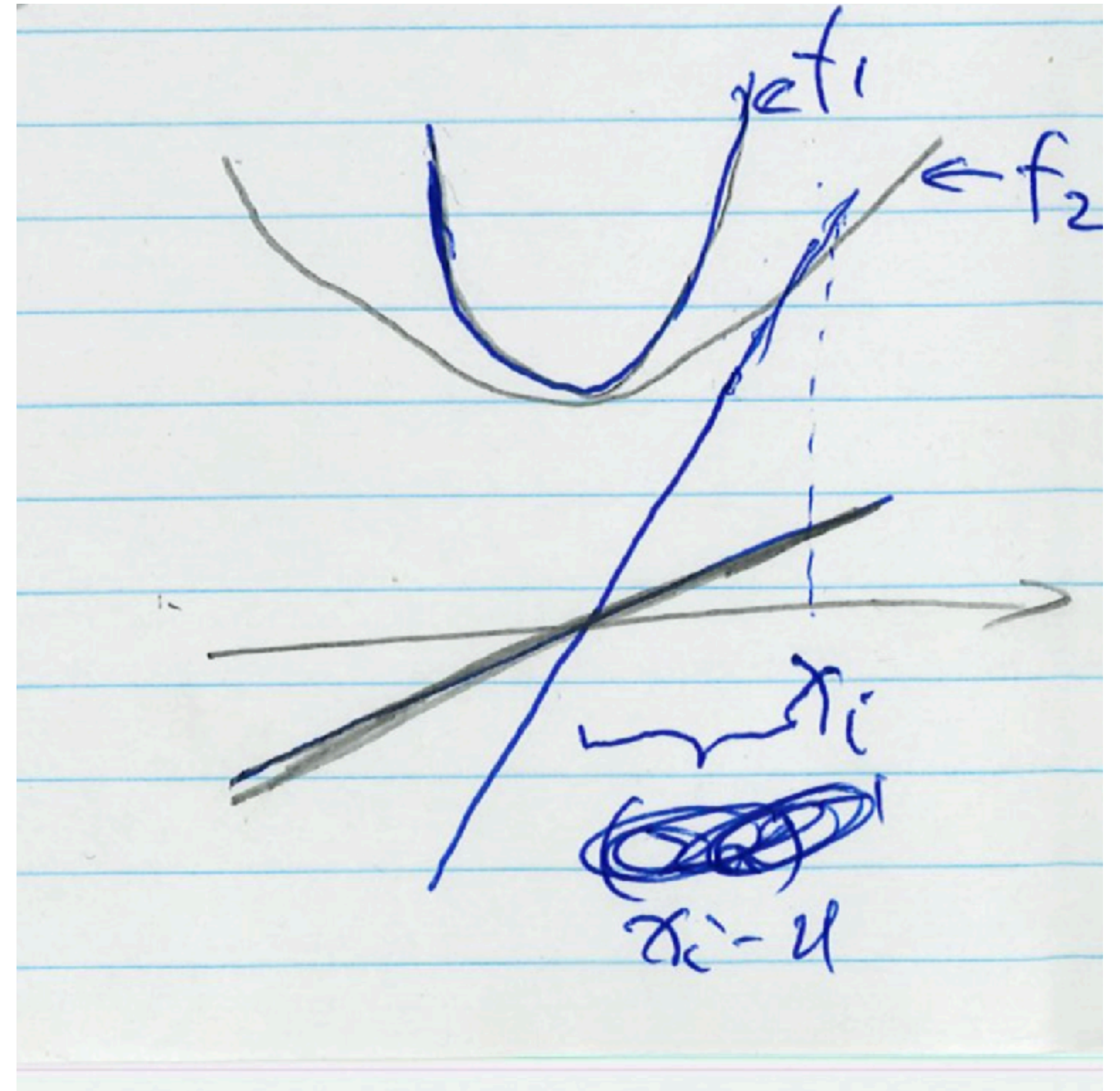
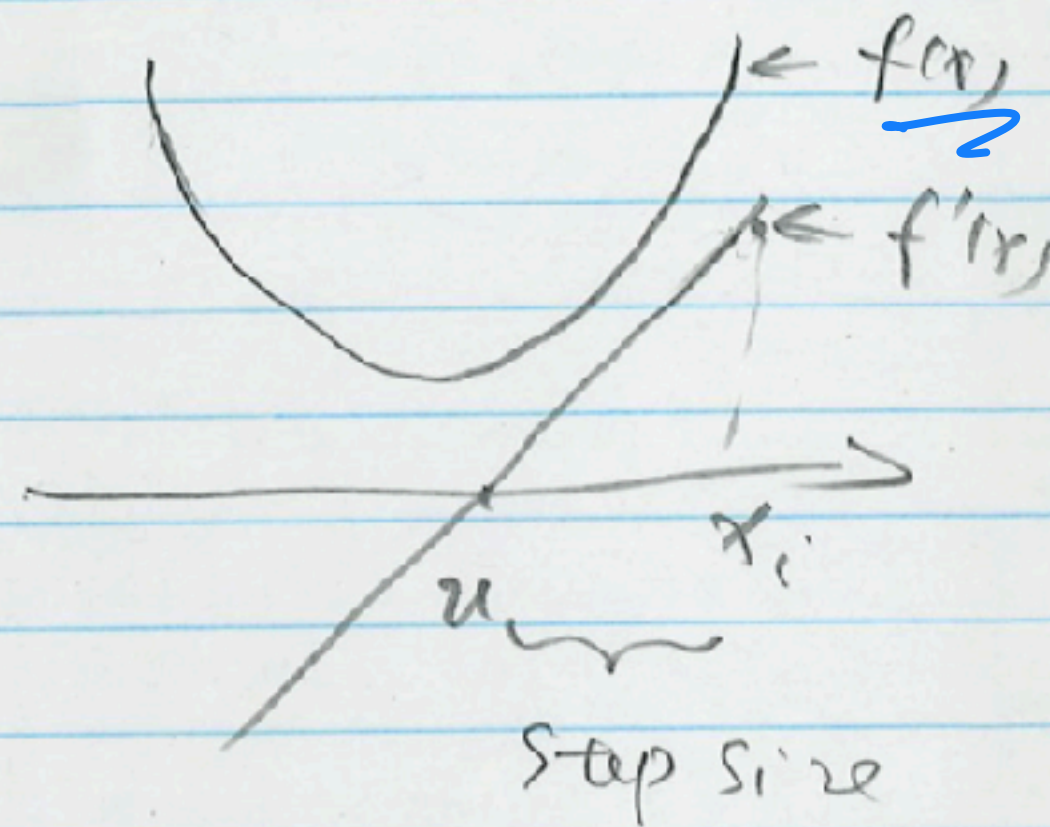
$$x_{i+1} = x_i - \frac{f'(x_i)}{f''(x_i)}$$

$$= x_i - \frac{2a(x_i - u)}{2a} = x_i - \underbrace{(x_i - u)}_{\text{step size}}$$

$[f''(x_i)]^{-1}$ is the ^{perfect} step size modifier

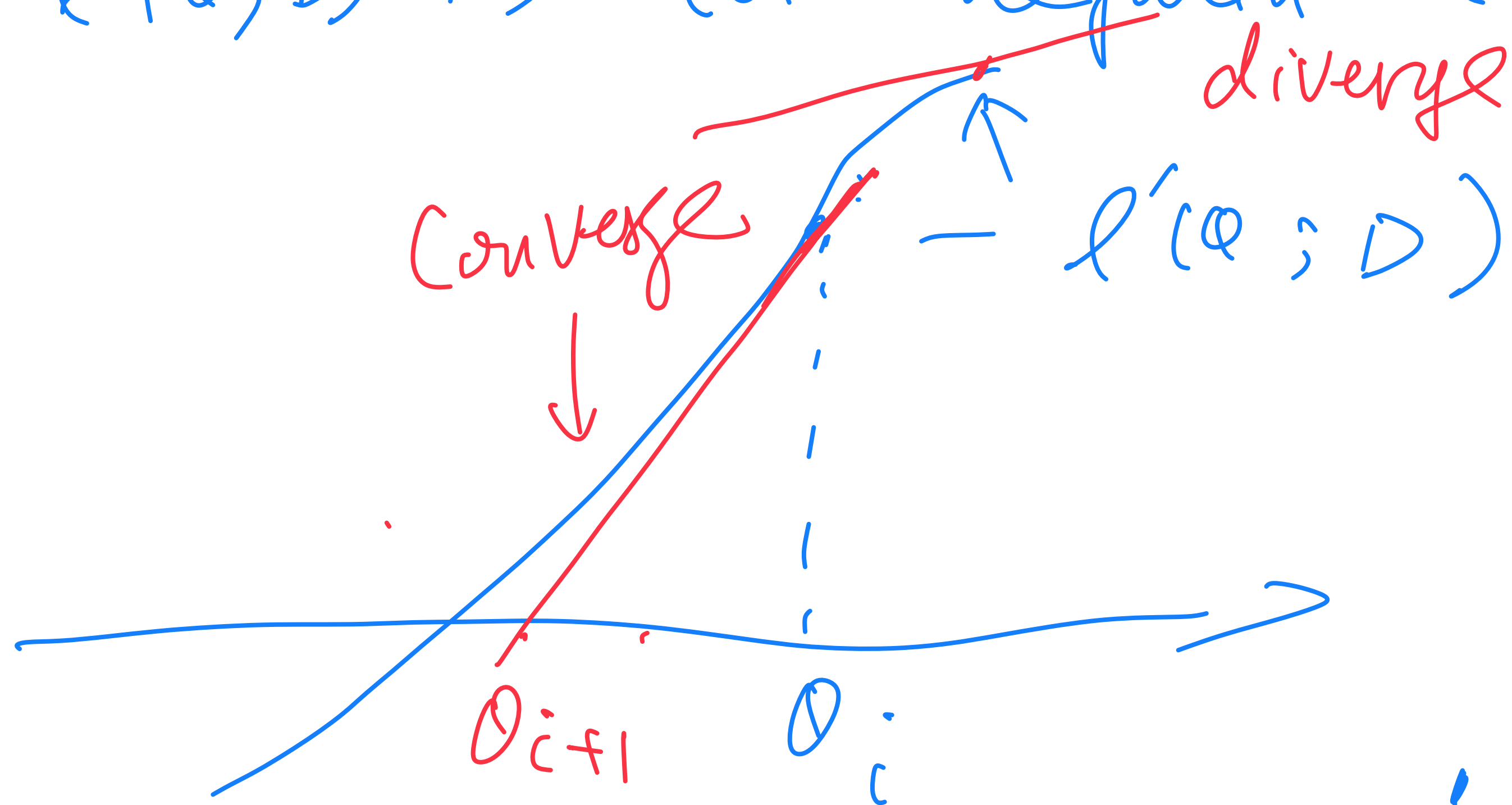
for x_i with slope $f'(x_i) = \text{~~2a(x_i - u)~~}$

if $f(x)$ is quadratic



Fisher Scoring

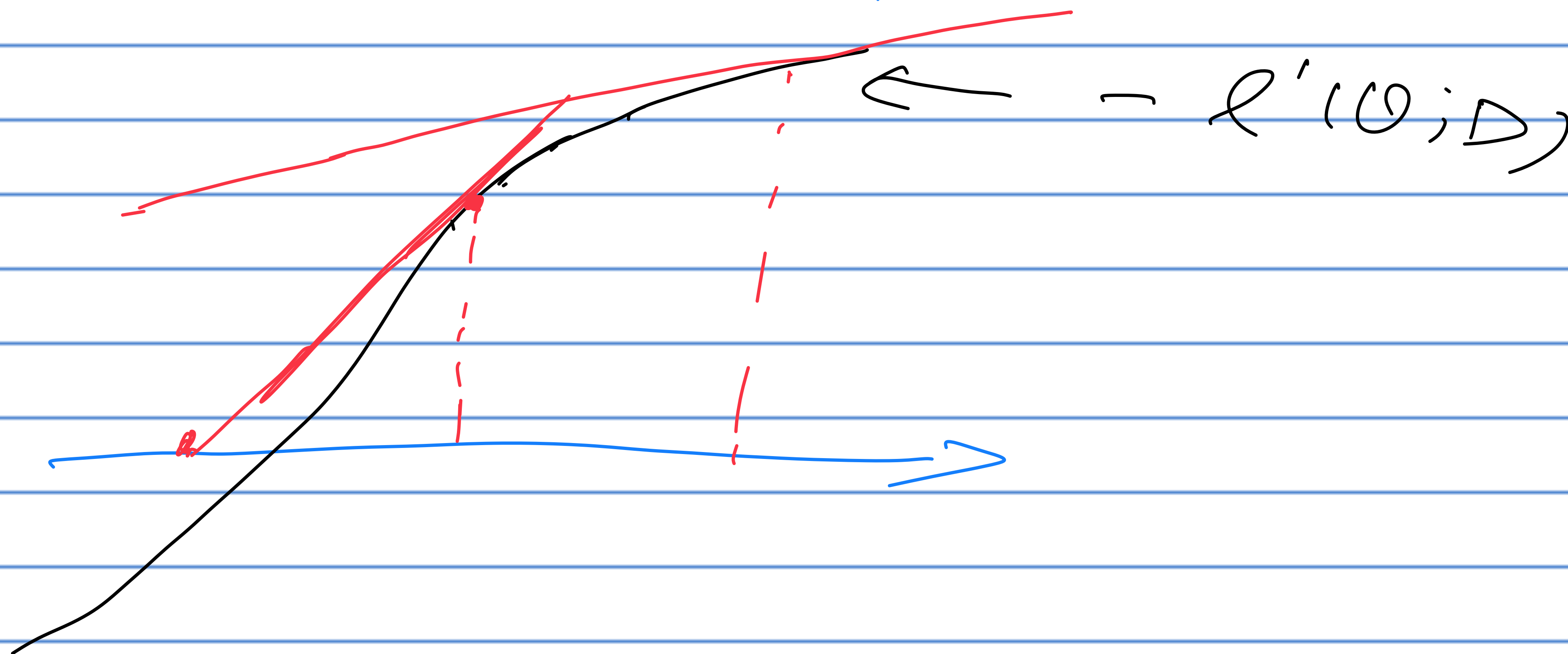
$-\ell(\theta; D)$ is the negative log-likelihood



Newton-Raphson

$$\theta_{i+1} = \theta_i - \frac{-\ell'(\theta_i; D)}{-\ell''(\theta_i; D)}$$

- $\ell''(\theta_i; D)$ is called
observed Fisher information at θ_i



Replacing $- \ell''(\theta_i; D)$ by the

Fisher information at θ_i :

$$n \cdot I(\theta_i) = \underset{\substack{\uparrow \\ \text{w.r.t. } D}}{\mathbb{E}_D} \left(- \ell''(\theta_i; D) \right)$$

$$\theta_{i+1} = \theta_i + \frac{+ \ell'(\theta_i; D)}{n \cdot I(\theta_i)}.$$

Exempl (Ordinary Poisson)

$$X_1, X_2, \dots, X_n \sim \text{pois}(\lambda)$$

$$\ell(\lambda) = \sum_{i=1}^n \log f(x_i; \lambda)$$

$$= \sum_{i=1}^n \log \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}$$

$$= \sum_{i=1}^n [-\lambda + x_i (\log \lambda) - (\log x_i!)]$$

$$= -n\lambda + \sum x_i \log \lambda + C$$

$$Q'(\lambda) = -n + \sum x_i / \lambda \stackrel{!}{=} 0$$

$$\lambda = \frac{\sum x_i}{n}$$

$$\lambda_{MLE} = \bar{x}$$

A demonstrating example (Truncated Poisson)

Suppose we only observe nonzero iid samples $X_1, \dots, X_n$ from a poisson distribution, whose probability function is:

$$P(x_1, \dots, x_n | \lambda) = \prod_{i=1}^n \frac{\exp(-\lambda) \lambda^{x_i}}{x_i!} \frac{1}{1 - \exp(-\lambda)} \quad (7.24)$$

$$P(x_i | \lambda) = \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} \cdot \frac{1}{P(x_i > 0)}$$

$$= \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} \cdot \frac{1}{1 - e^{-\lambda}}$$

$$l(\lambda) = -\lambda n + \sum x_i \log(\lambda) - n \log(1 - \exp(-\lambda)) \quad (7.25)$$

The score function $s(\lambda) = l'(\lambda)$:

$$s(\lambda) = \frac{\sum x_i}{\lambda} - \frac{n}{1 - \exp(-\lambda)} \quad (7.26)$$

The equation $s(\lambda) = 0$ is equivalent to:

$$\frac{\bar{x}}{\lambda} = \frac{1}{1 - e^{-\lambda}}$$

$$\lambda = \bar{x} (1 - \exp(-\lambda))$$

Simple iteration (7.27)

The derivative of score function:

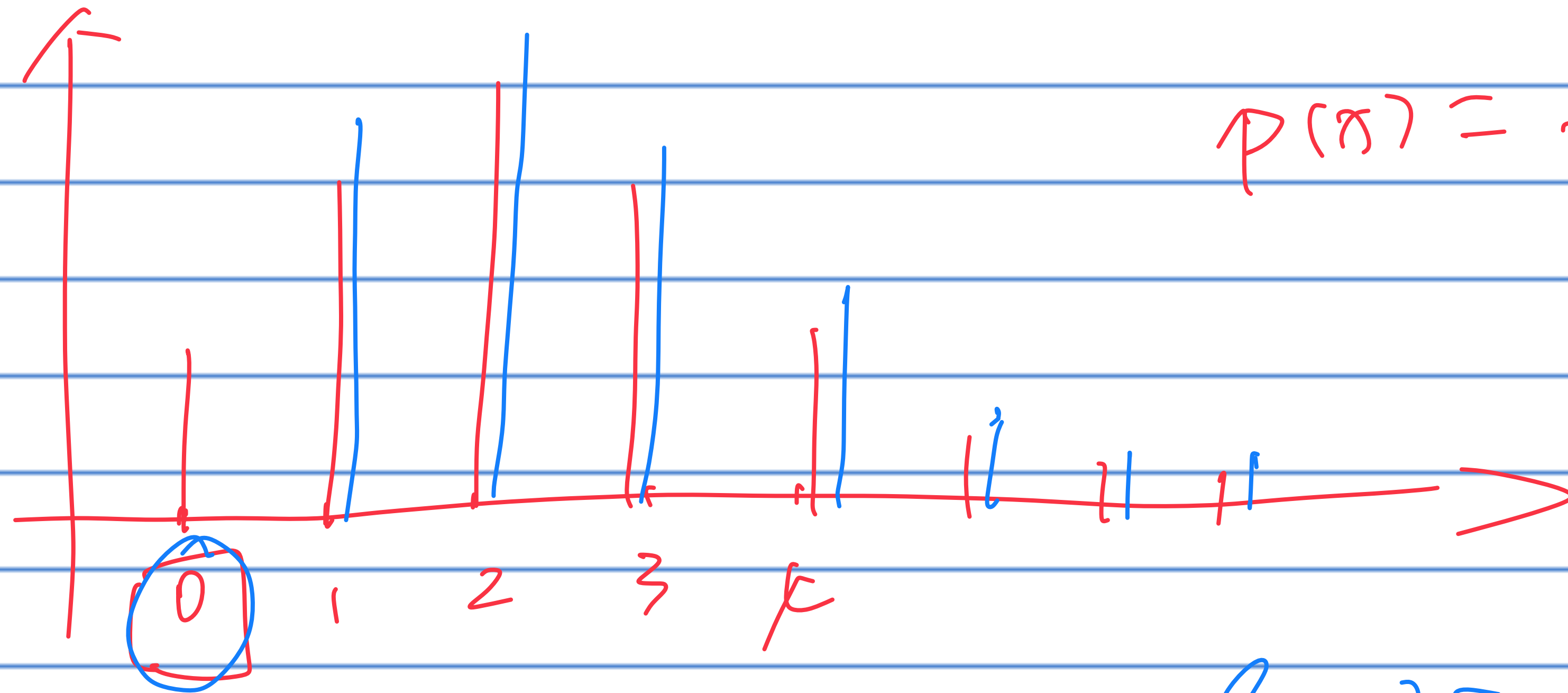
$$s'(\lambda) = -\frac{\sum x_i}{\lambda^2} + n \frac{\exp(-\lambda)}{(1 - \exp(-\lambda))^2} \quad (7.28)$$

The fisher information based on n sample is:

$$E(x_i) = \lambda \cdot \frac{1}{1 - e^{-\lambda}}$$

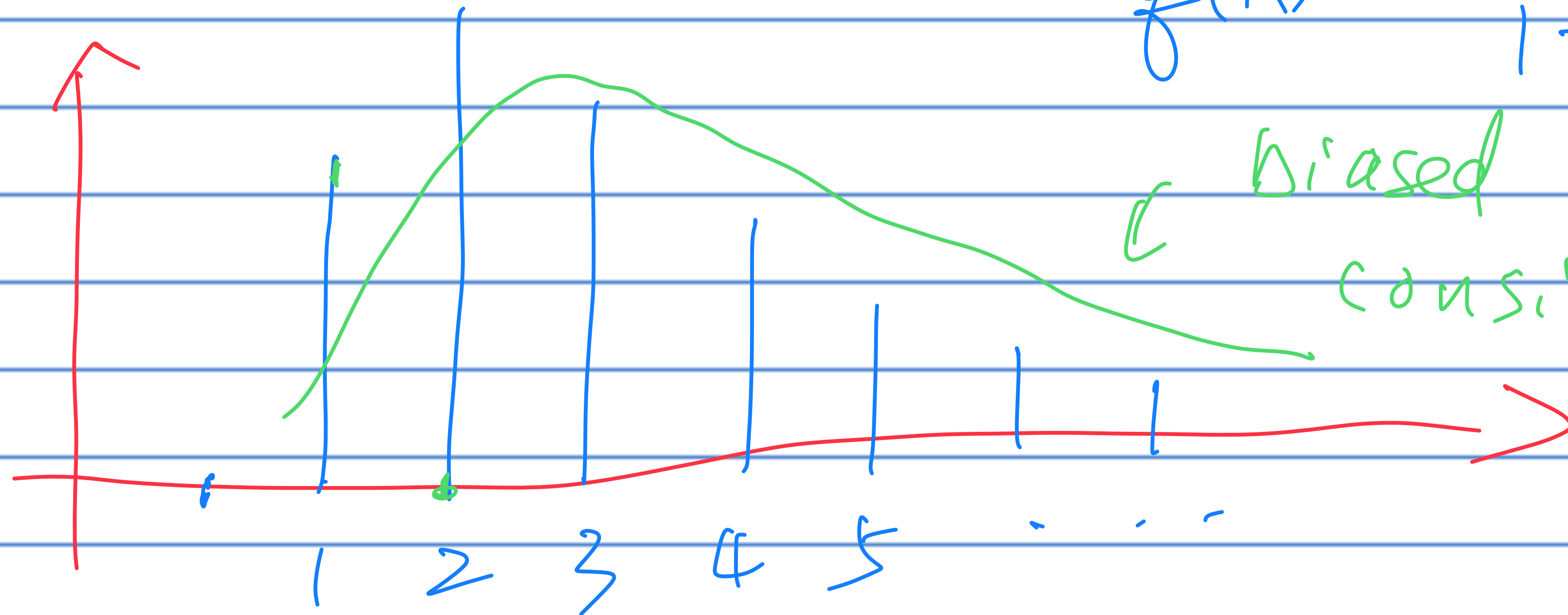
$$n I(\lambda) = -E(s'(\lambda)) \quad (7.29)$$

$$= \frac{1}{\lambda(1 - \exp(-\lambda))} - n \frac{\exp(-\lambda)}{(1 - \exp(-\lambda))^2} \quad (7.30)$$



$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$q(x) = \frac{p(x)}{1 - e^{-\lambda}}$$



Biased without
considering zero
truncation

$$Y \sim \text{Pois}(\lambda)$$

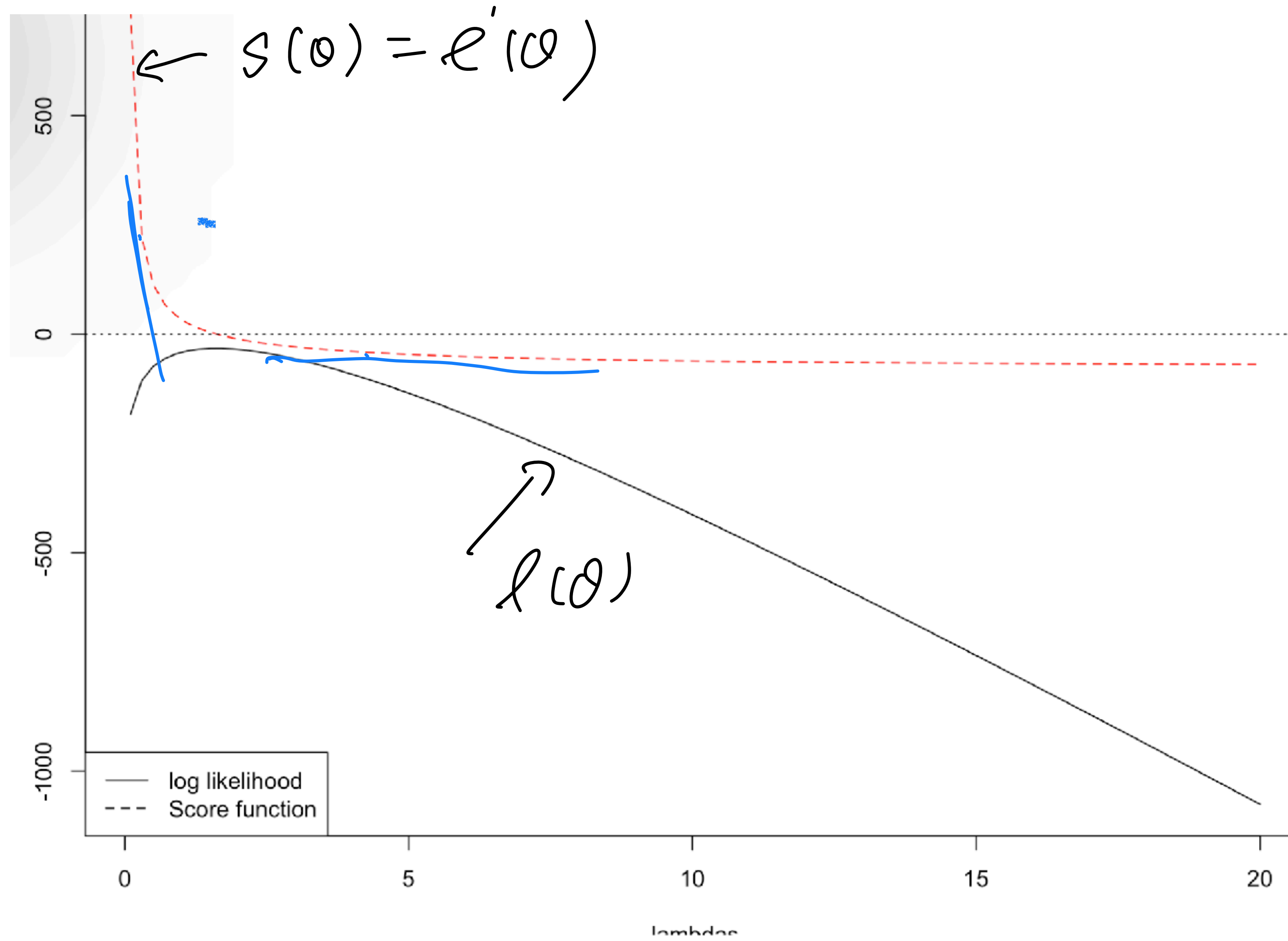
$$E(Y) = \sum_{y=0}^{\infty} \frac{e^{-\lambda} \lambda^y}{y!} \cdot y = \lambda$$

$$X \sim Y | Y > 0$$

$$E(X) = \sum_{x=1}^{\infty} \frac{e^{-\lambda} \lambda^x}{x!} x \cdot \left(\frac{1}{1 - e^{-\lambda}} \right)$$

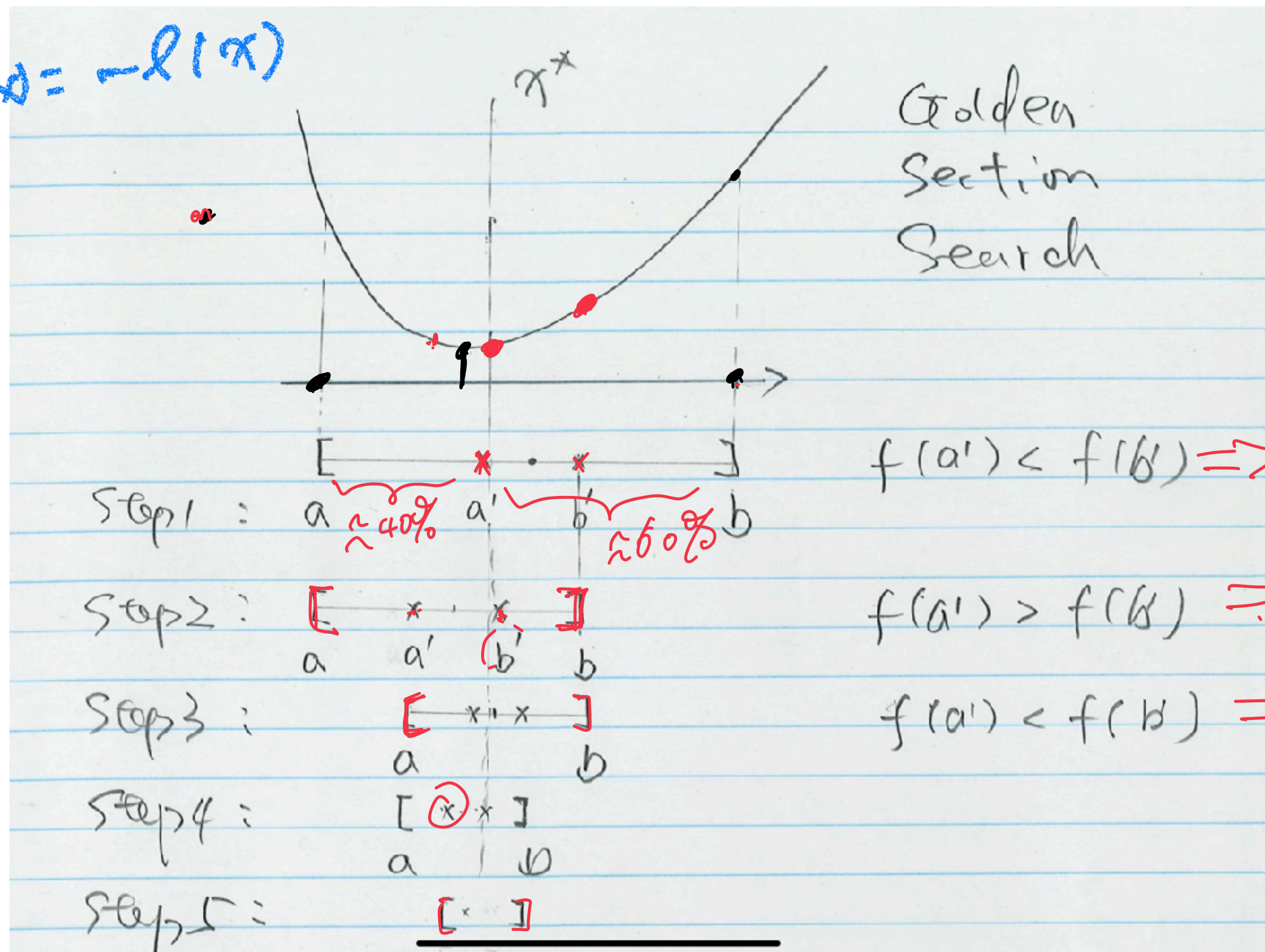
$$= \left(\lambda \right) \frac{1}{1 - e^{-\lambda}}$$

as the first term in $E(Y)$ is 0



Golden Section Method.

$$f(x) = -\ln(x)$$



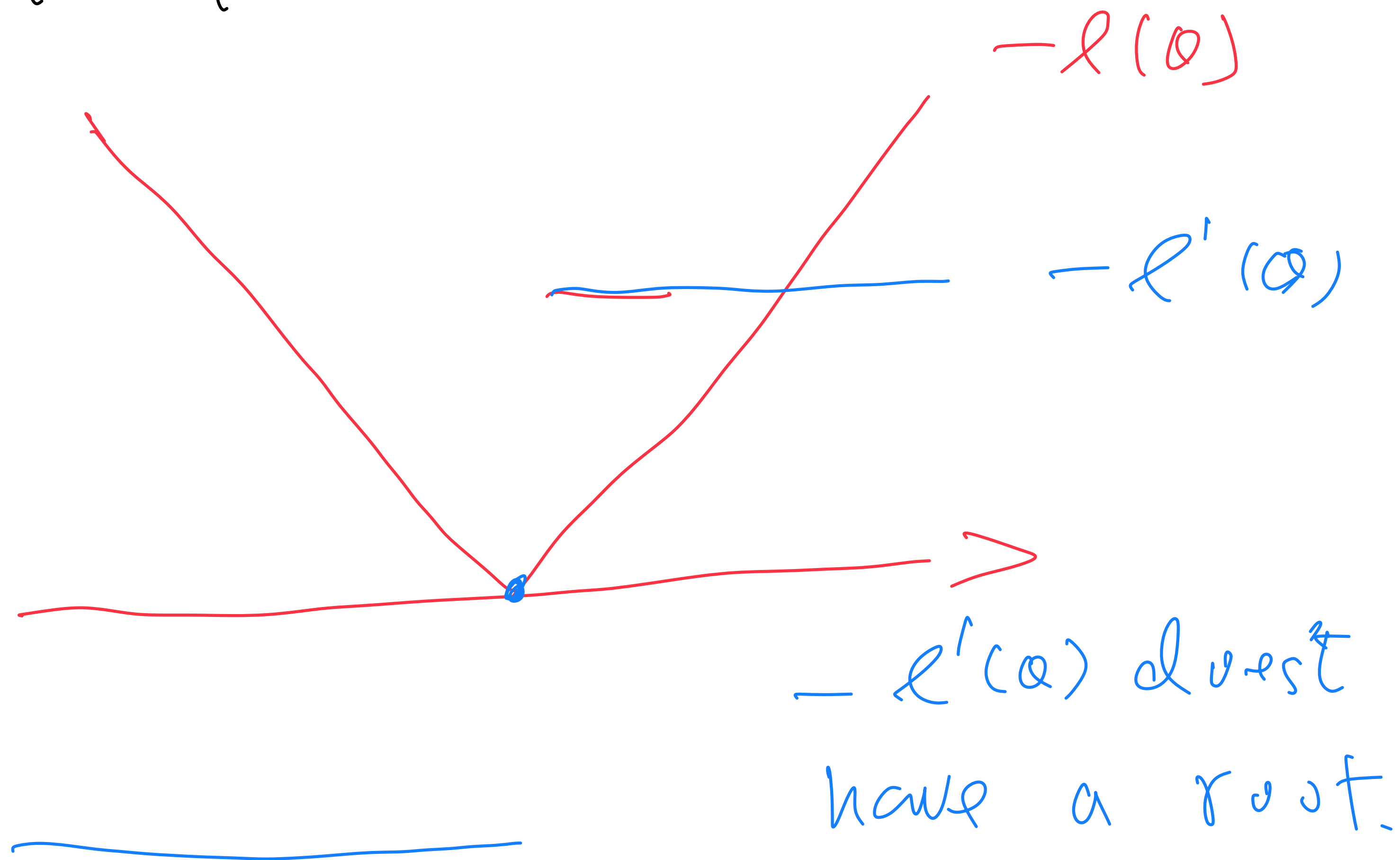
$$f(a') < f(b) \Rightarrow b = b'$$

$$f(a') > f(b) \Rightarrow a = a'$$

$$f(a') < f(b) \Rightarrow b = b'$$

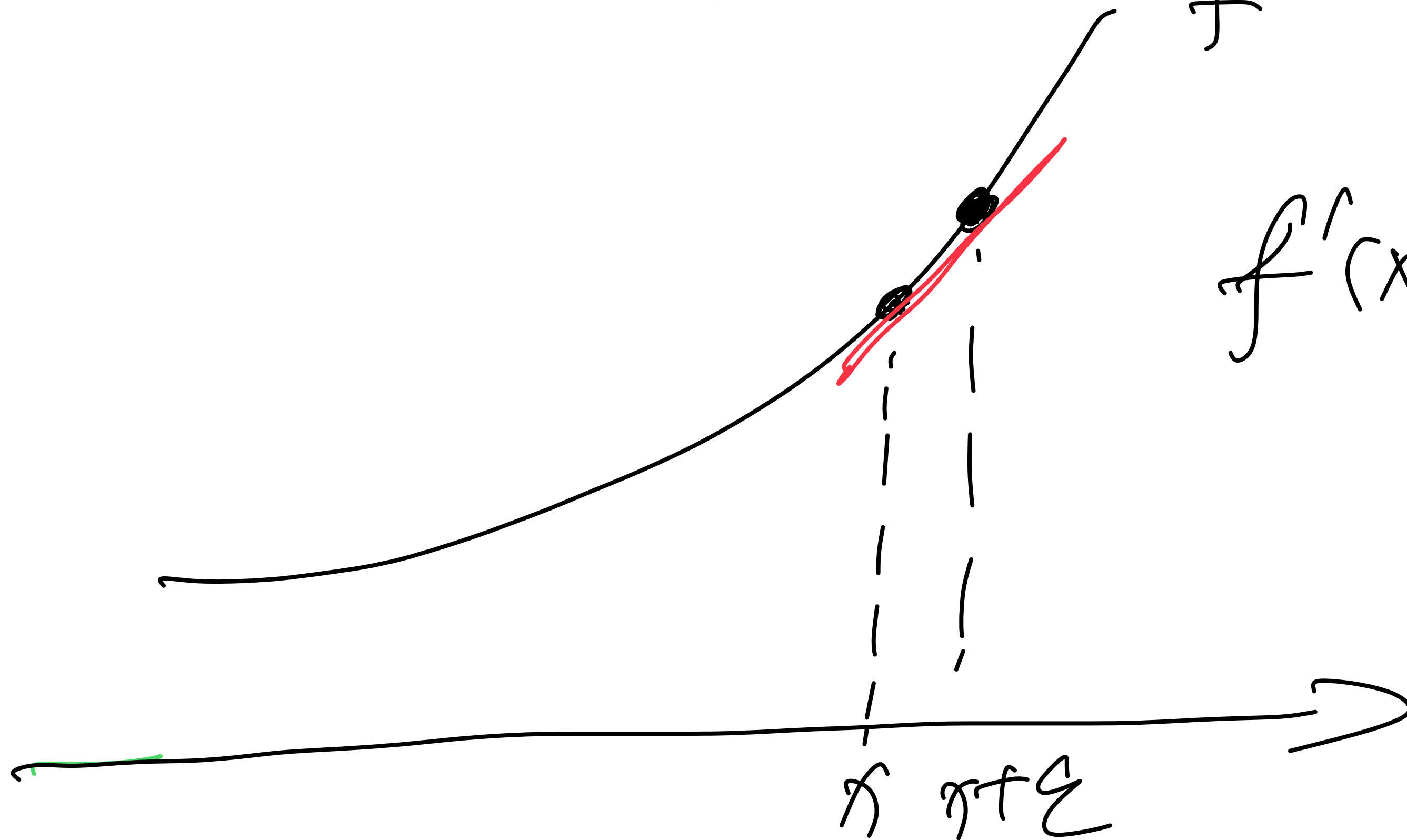
Shrink the side with larger $f(a')$ or $f(b')$.

An example for Golden Section Search



Newton-Raphson Method will fail for
such problems

Numerically evaluate derivative f

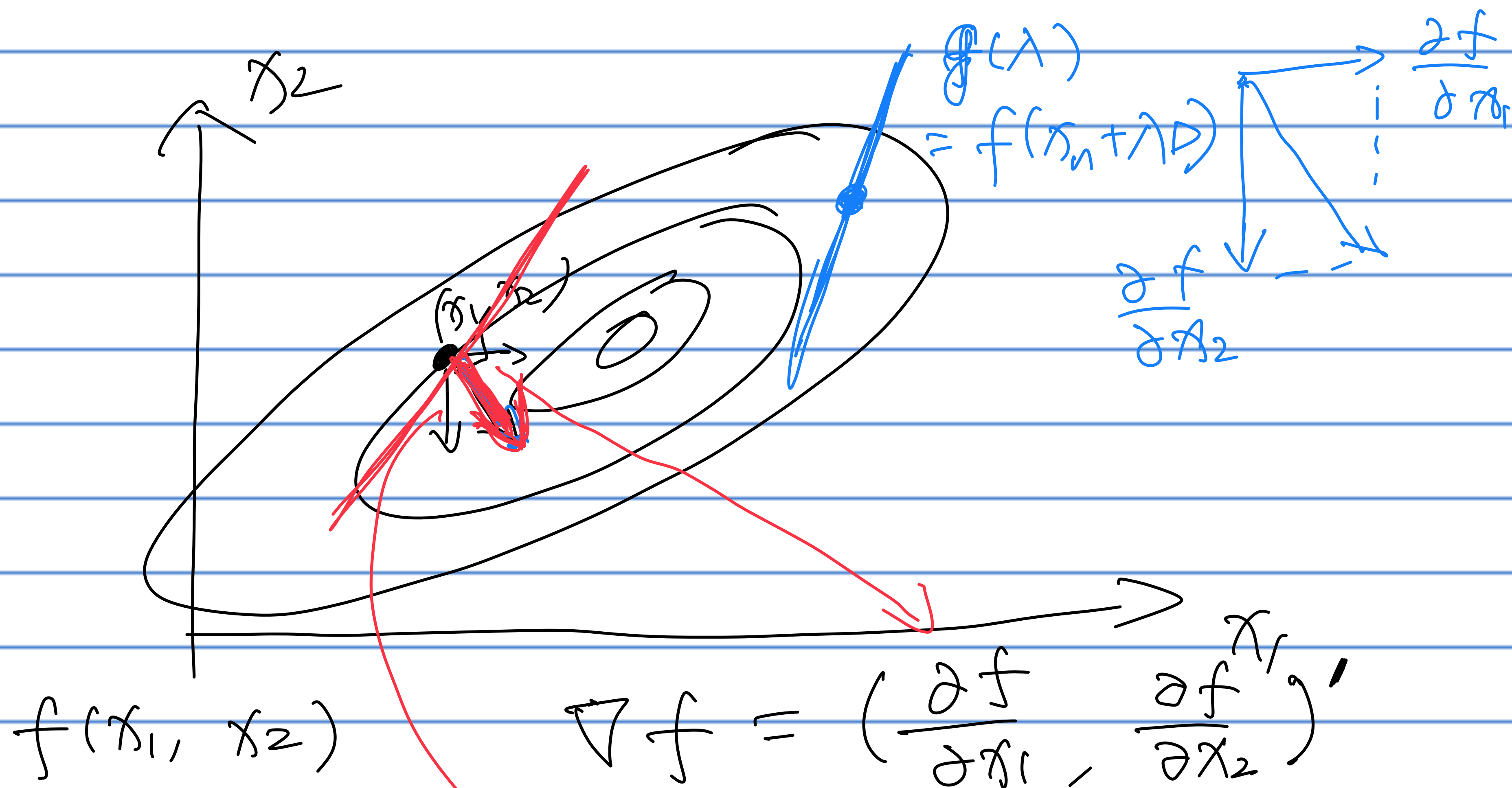


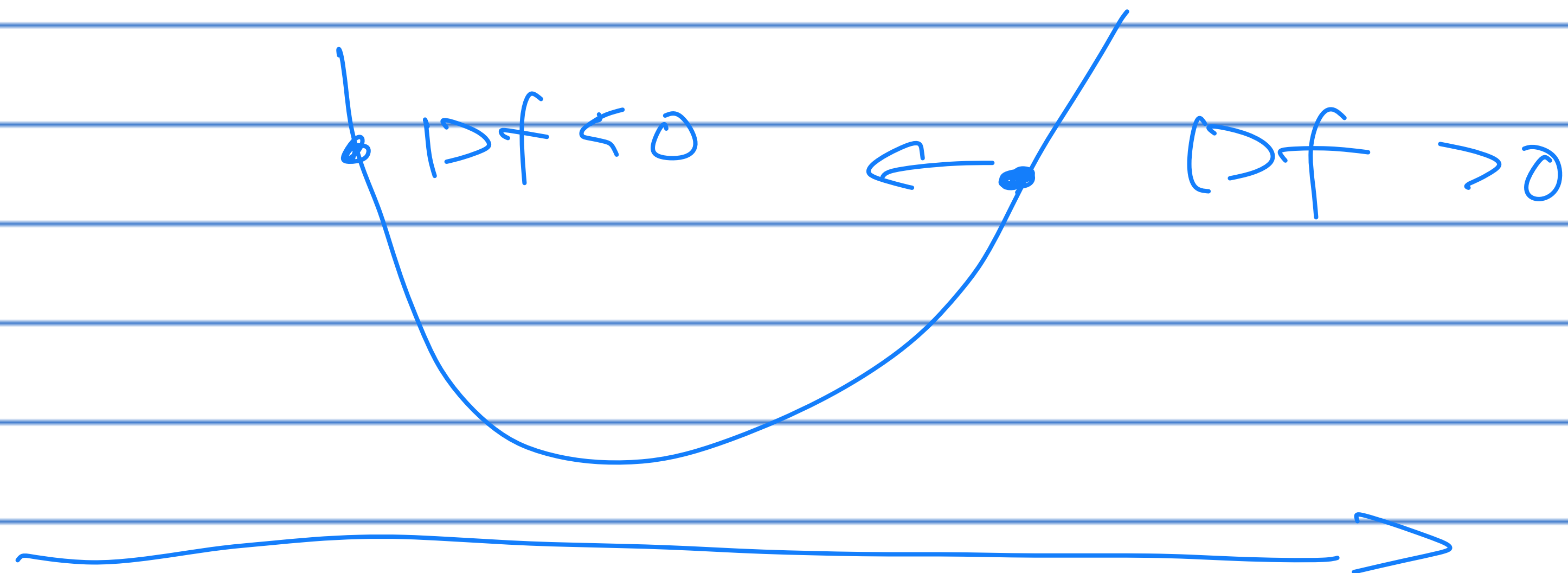
$$f'(x) \approx$$

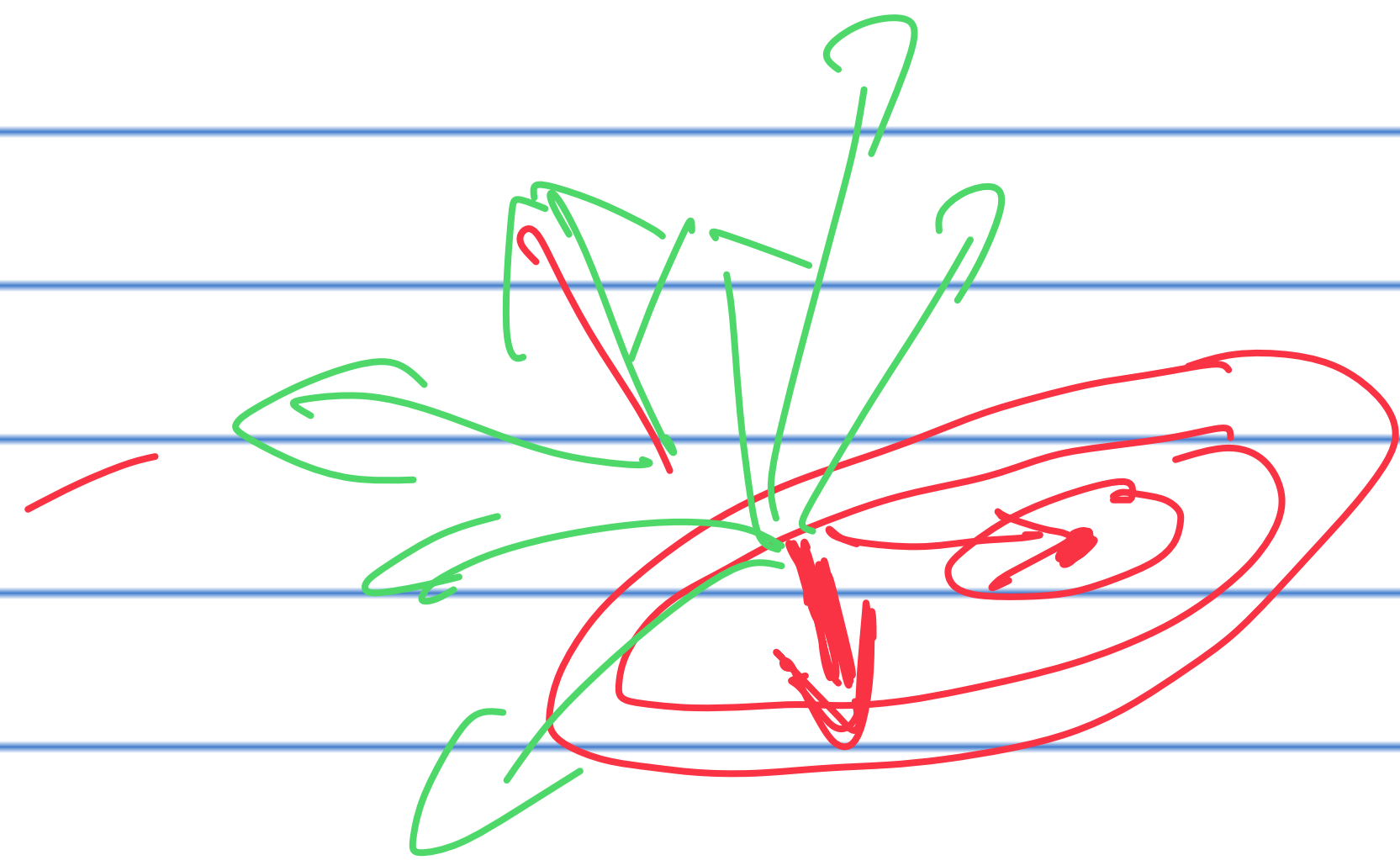
$$\frac{f(x+\epsilon) - f(x)}{\epsilon}$$

$\lim$

Multivariate Optimization







Hessian

$$\nabla^2 f = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_p} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2 \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_2 \partial x_p} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_p \partial x_1} & \frac{\partial^2 f}{\partial x_p \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_p \partial x_p} \end{pmatrix}_{p \times p}$$

$$x = (x_1, x_2, \dots, x_p)$$

$$(x-u)' A (x-u) + C,$$

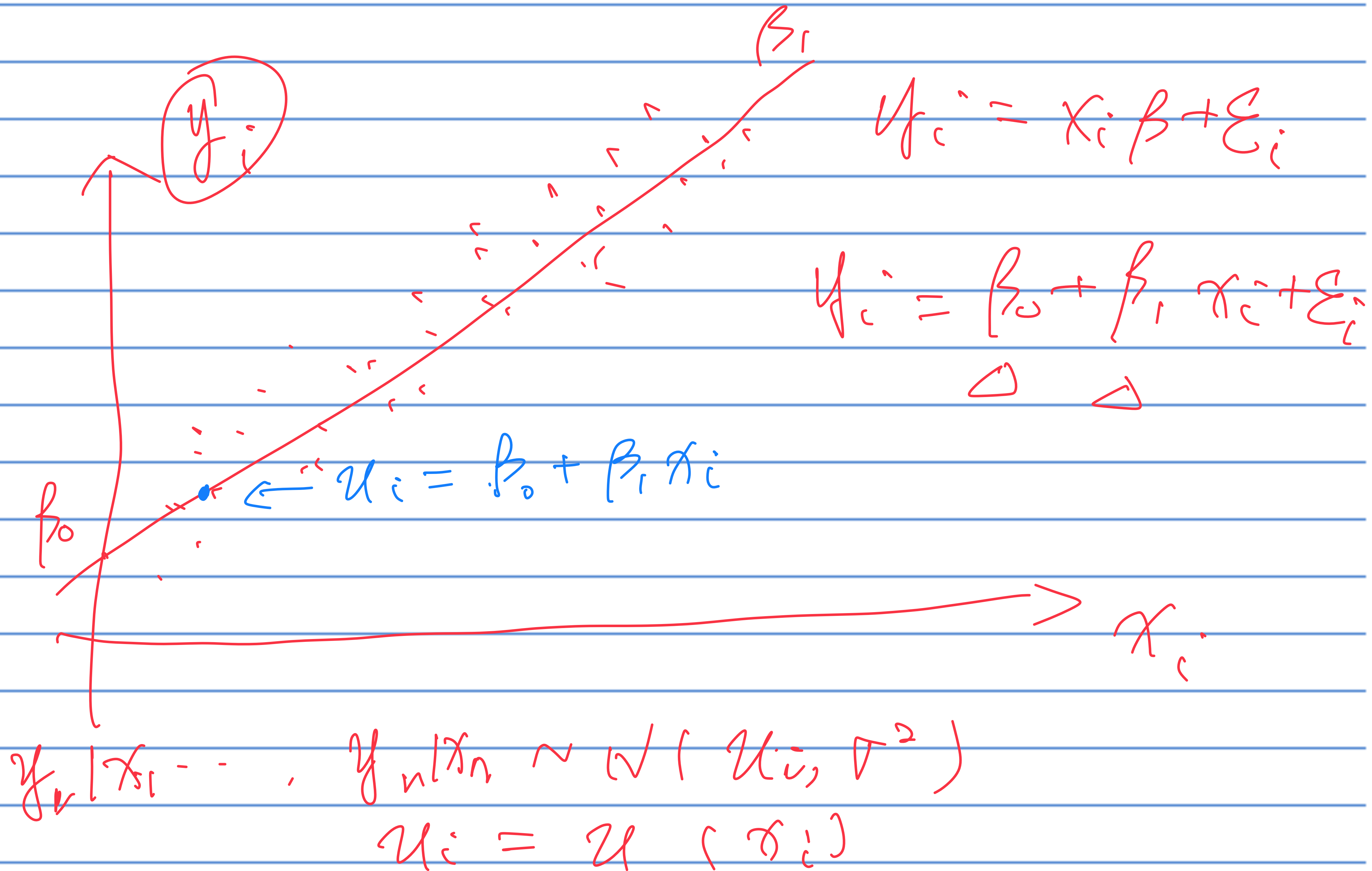
$$ax^2 + bx + c$$

N-R iteration

$$x_{n+1} = x_n - \nabla^2 f(x_n)^{-1} \nabla f(x_n)$$

$p \times 1$ $p \times p$ $p \times 1$

$$x_{n+1} = x_n - \frac{S(x_n)}{S'(x_n)}$$



Bernoulli R.V.

$$y_1, y_2, \dots, y_n \sim \text{Bern}(p)$$

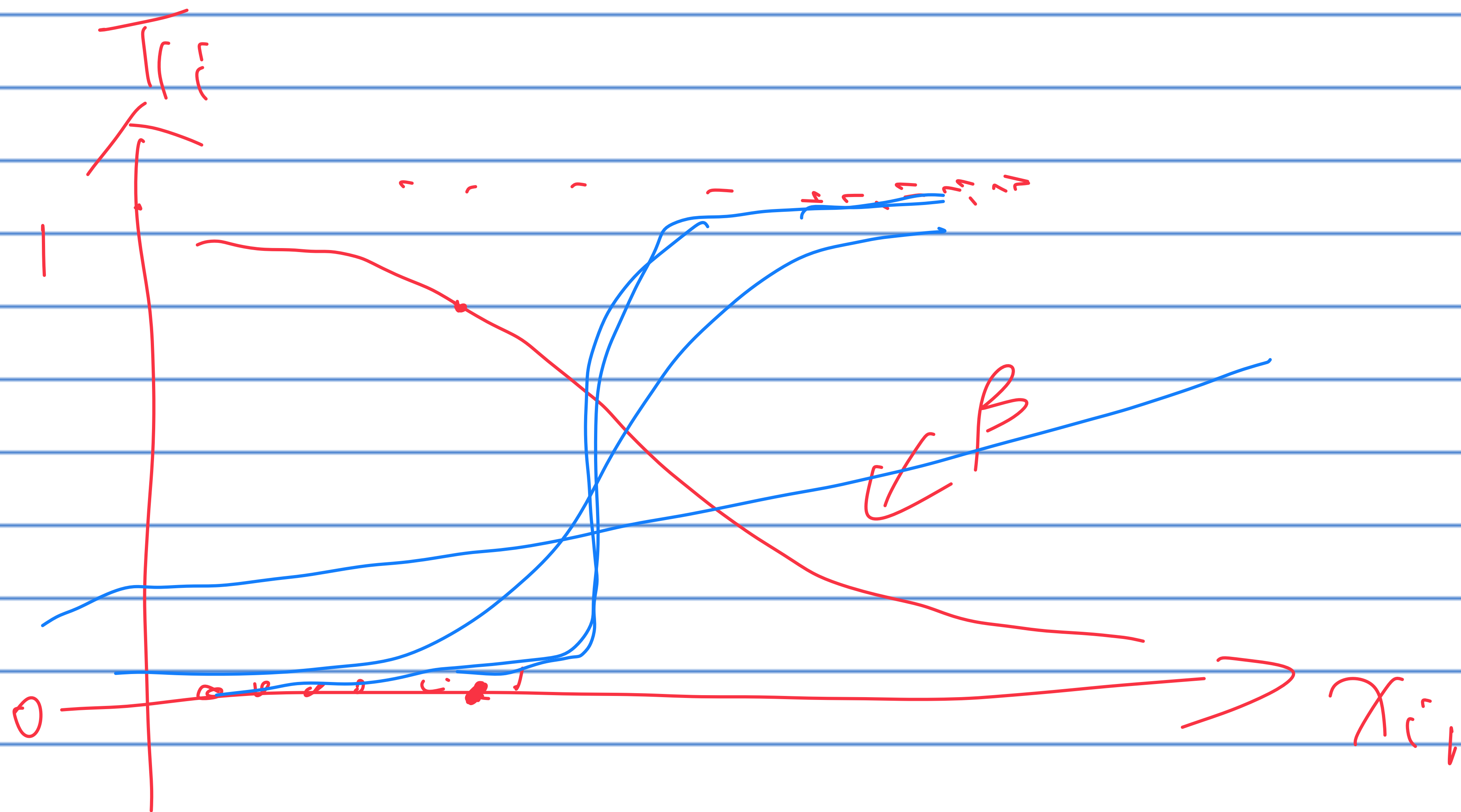
$$p(y_i) = p^{y_i} (1-p)^{1-y_i}$$

$$= \begin{cases} p, & \text{if } y_i = 1 \\ 1-p, & \text{if } y_i = 0 \end{cases}$$

Logistic Regression

$$y_i | x_i \sim \text{Bern}(\pi_i), \quad \pi_i = \pi(x_i)$$





$$\pi_i = \pi(x_i) = \frac{\exp(\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip})}{1 + \exp(\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip})}$$

$$\frac{e^x}{1+e^x} \quad //$$

$$\frac{1}{1-e^{-x}}$$

Likelihood function

$$f(y_1, \dots, y_n | x_1, \dots, x_n, \beta)$$

$$= \prod_{i=1}^n f(y_i | x_i, \beta)$$

$$= \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1 - y_i}$$

$$= \prod_{i=1}^n \left(\frac{e^{x_i \beta}}{1 + e^{x_i \beta}} \right)^{y_i} \left(\frac{1}{1 + e^{x_i \beta}} \right)^{1 - y_i}$$

$$= \frac{e^{\sum_{i=1}^n y_i x_i \beta}}{\sum_{i=1}^n (1 + e^{x_i \beta})}$$

$$\sum_{i=1}^n (1 + e^{x_i \beta})$$

$$Q(\beta; y, x) = \sum_{i=1}^n y_i (\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}) - \sum_{i=1}^n \log(1 + e^{\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}})$$

$$\frac{\partial \ell}{\partial \beta_0} = \sum_{i=1}^n y_i - \sum_{i=1}^n \frac{e^{x_i \beta_0}}{1 + e^{x_i \beta_0}}$$

$$= \sum_{i=1}^n (y_i - \pi_i)$$

$$\frac{\partial \ell}{\partial \beta_1} = \sum_{i=1}^n y_i x_{i1} - \sum_{i=1}^n \frac{e^{x_i \beta_0} x_{i1}}{1 + e^{x_i \beta_0}}$$

$$= \sum_{i=1}^n (y_i - \pi_i) x_{i1}$$



H isn't

positive-definite

$$\nabla \ell =$$

$$\begin{bmatrix} \frac{\partial \ell}{\partial \beta_0} \\ \frac{\partial \ell}{\partial \beta_1} \\ \vdots \\ \frac{\partial \ell}{\partial \beta_p} \end{bmatrix}$$

$$\nabla^2 \ell //$$

$$\frac{\partial^2 \ell}{\partial \beta_0 \partial \beta_0}$$

...

$$\frac{\partial^2 \ell}{\partial \beta_0 \partial \beta_p}$$

$$\frac{\partial^2 \ell}{\partial \beta_p \partial \beta_0}$$

...

$$\frac{\partial^2 \ell}{\partial \beta_p \partial \beta_p}$$

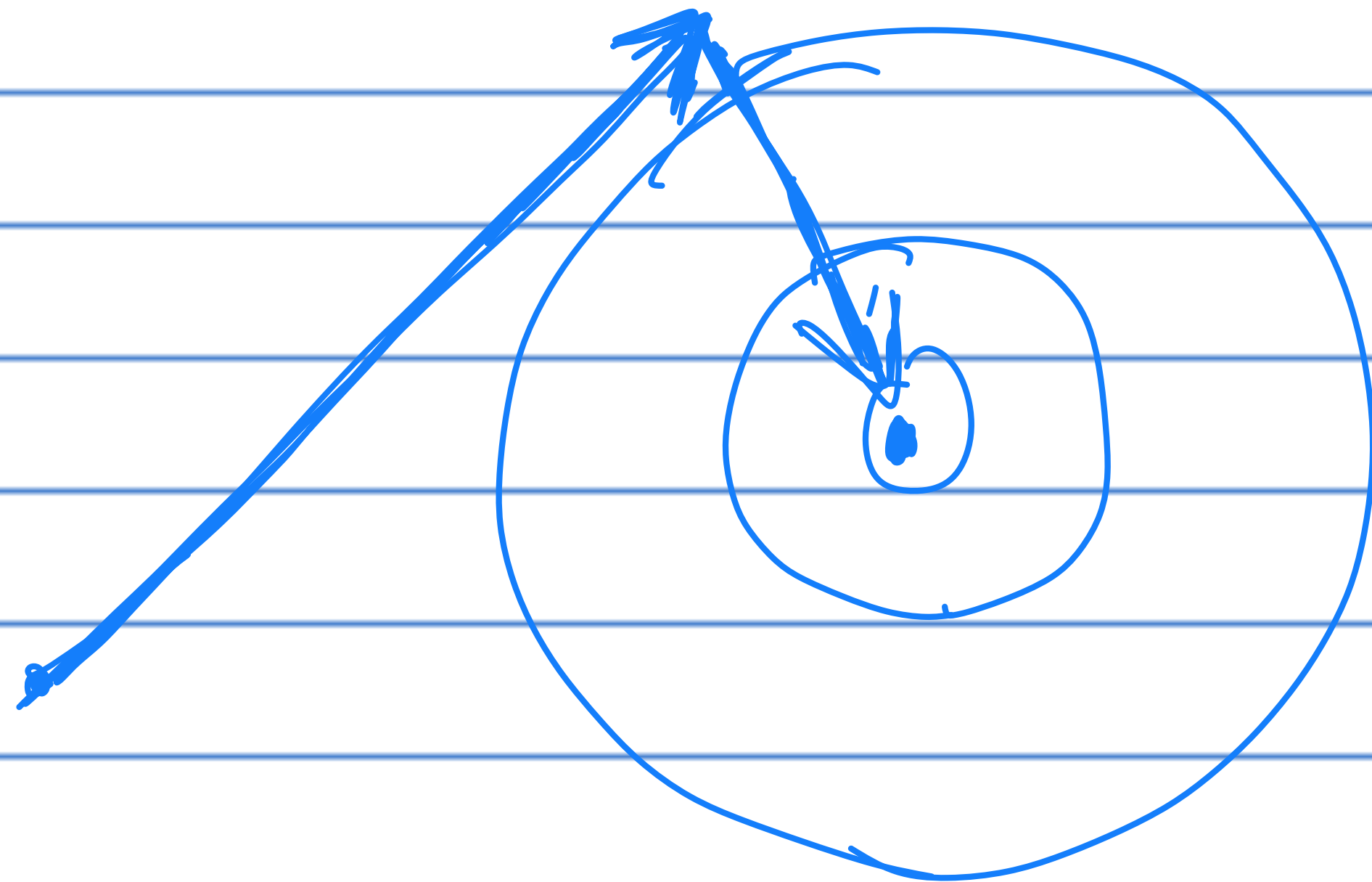
Variance for $\hat{\beta}$:

$$V(\hat{\beta}) = [n \cdot I(\beta)]^{-1}$$

$$\approx \left[-\nabla^2 \ell(\hat{\beta}) \right]^{-1}$$

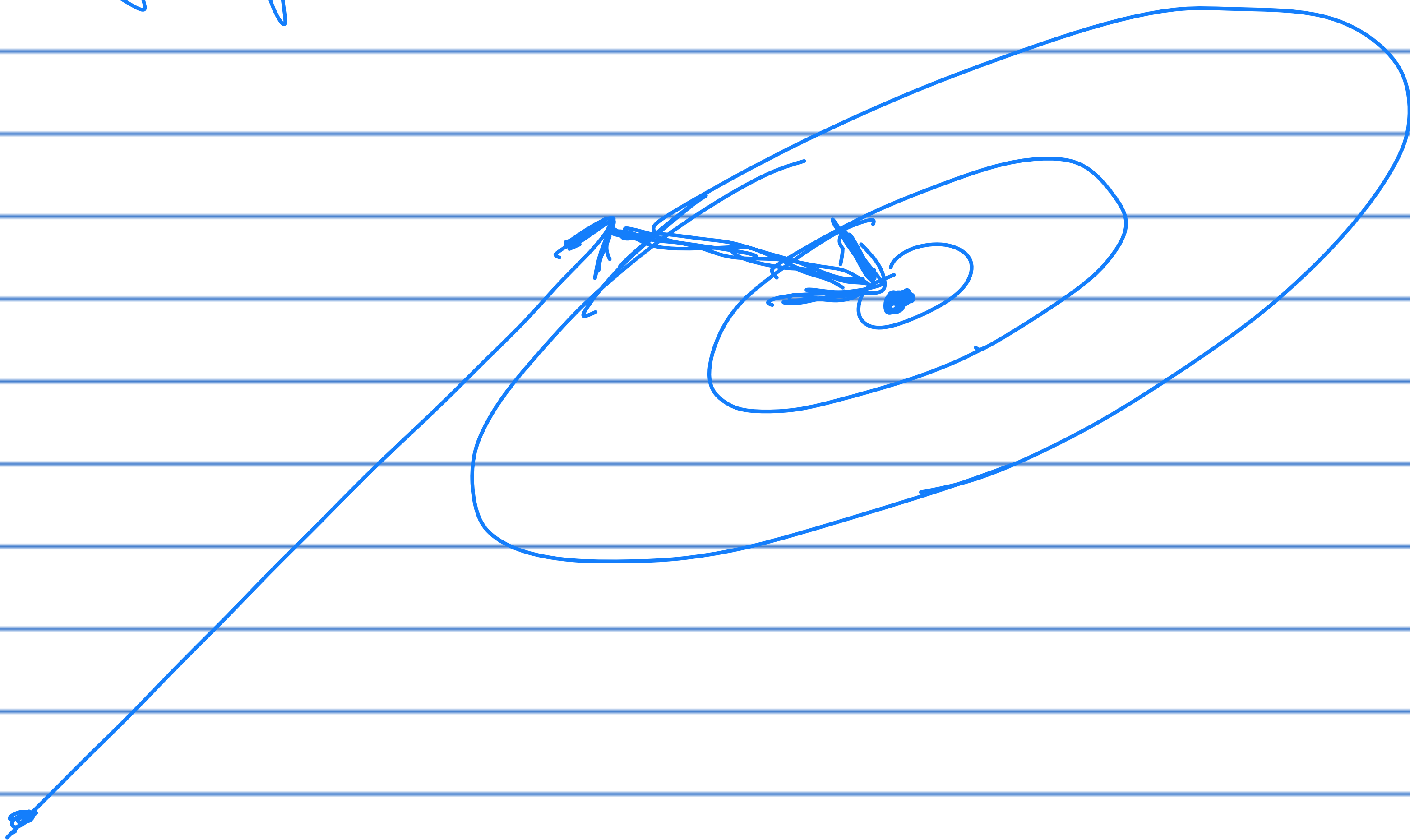
$$V(\hat{\beta}) = \begin{pmatrix} \hat{\sigma}_{11}^2 & \hat{\sigma}_{12} & \dots & \hat{\sigma}_{1p} \\ \hat{\sigma}_{p1} & \hat{\sigma}_{p2} & \dots & \hat{\sigma}_{pp}^2 \end{pmatrix}, \quad \sqrt{\text{diag}(V(\hat{\beta}))} = \hat{SD}(\hat{\beta})$$

$$f(x, y) = x^2 + y^2$$



Q-Conjugate

$$f(x) = x'Qx$$



Neural Network

